MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

49PR COMPACT WORKSHOP & WAREHOUSE

DESIGN CALCULATION OF FOUNDATION

AS BUILT

MD-216-49PR-EG-CV-CAL-1625

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
D01	25-10-2023	AS BUILT	SHD	SGC	PRI	
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A01	06-05-2020	FIRST ISSUE	SHD	SGC	PRI	

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01			
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2014	Building Code Requirement of Structural Concrete

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
LRFD	Load and Resistance Factor Design
F.O.S	Factor of Safety

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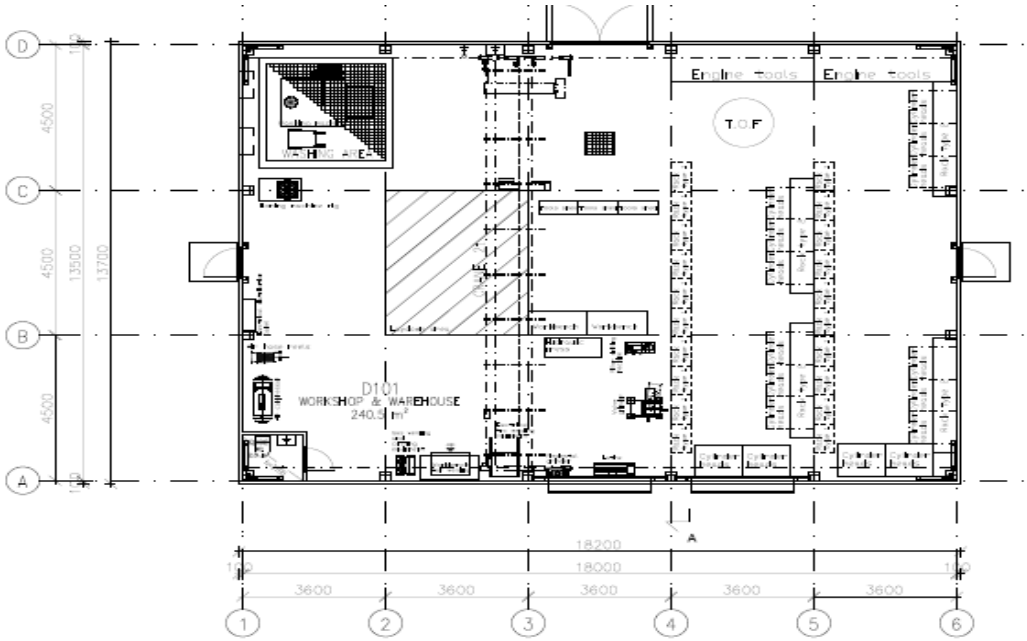
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1. INTENT

The document is prepared for Design Calculation of Column Foundation for **WORKSHOP & WAREHOUSE**

2. SCOPE

The document is prepared for Design Calculation of Column Foundation for Workshop & Warehouse
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
The foundation is designed for wind load as seismic load is less than wind load.(Refer MD-216-49PR-EG-ST-CAL-2420 for Workshop structural design calculation)
Design of Reinforced Concrete Foundations followed Strength Design as per SBC 304:2018 & ACI 318:2008
Net Safe Bearing Capacity is obtained from Geotechnical report BH28 . Approximate value of 50T/sq.m is considered



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3. Design of Workshop & Warehouse Foundation

3.1 Design of Foundation mkd. F1 (GRID D1,4,5,6)

Load Detail	D(1,4,5,6)		
	FX (T)	FY (T)	FZ (T)
1 DL	1	6	0
2 LLROOF	1	3	0
3 CL1(L+XS)	0	2	0
4 CL2(L-XS)	0	2	0
5 CL3(R+XS)	1	5	0
6 CL4(R-XS)	0	5	0
7 CL5(L+ZT)	0	2	0
8 CL6(L-ZT)	0	2	0
9 CL7(R+ZT)	0	5	0
10 CL8(R-ZT)	0	5	0
21 WL1(+X)S	0	-1	0
22 WL2(+X)P	0	-2	0
23 WL3(-X)S	-1	-2	0
24 WL4(-X)P	-1	-3	0
25 WL5(+Z)S	0	-1	0
26 WL6(+Z)P	0	-3	0
27 WL7(-Z)S	0	-2	0
28 WL8(-Z)P	0	-3	0

STRENGTH DESIGN	D(1,4,5,6)		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4D	1.41	8.54	-0.01
1.4D+1.7(CL1)+0.5Lr	2.4	13.6	0.0
1.4D+1.7(CL2)+0.5Lr	2.3	13.5	0.0
1.4D+1.7(CL3)+0.5Lr	2.9	18.3	0.0
1.4D+1.7(CL4)+0.5Lr	2.3	18.2	0.0
1.4D+1.7(CL5)+0.5Lr	2.3	13.5	0.0
1.4D+1.7(CL6)+0.5Lr	2.3	13.5	0.0
1.4D+1.7(CL5)+0.5Lr	2.6	18.2	0.0
1.4D+1.7(CL6)+0.5Lr	2.6	18.2	0.0
1.2D+1.6Lr+1.0(CL1)	3.3	14.4	0.0
1.2D+1.6Lr+1.0(CL2)	3.2	14.4	0.0
1.2D+1.6Lr+1.0(CL3)	3.6	17.2	0.0
1.2D+1.6Lr+1.0(CL4)	3.3	17.1	0.0
1.2D+1.6Lr+1.0(CL5)	3.3	14.4	0.0
1.2D+1.6Lr+1.0(CL6)	3.3	14.4	0.0
1.2D+1.6Lr+1.0(CL7)	3.4	17.2	0.0
1.2D+1.6Lr+1.0(CL8)	3.4	17.2	0.0
1.2D + 1.6WL1+1.0L+0.5Lr	1.7	7.6	0.0
1.2D + 1.6WL2+1.0L+0.5Lr	1.9	6.0	0.0
1.2D + 1.6WL3+1.0L+0.5Lr	-0.5	6.3	0.0
1.2D + 1.6WL4+1.0L+0.5Lr	-0.3	4.9	0.0
1.2D + 1.6WL5+1.0L+0.5Lr	1.5	6.5	0.0
1.2D + 1.6WL6+1.0L+0.5Lr	1.6	4.9	0.0
1.2D + 1.6WL7+1.0L+0.5Lr	1.5	6.5	0.0
1.2D + 1.6WL8+1.0L+0.5Lr	1.7	4.9	0.0
0.9D+1.6WL1	0.8	4.2	0.0
0.9D+1.6WL2	1.0	2.6	0.0
0.9D+1.6WL3	-1.4	2.9	0.0
0.9D+1.6WL4	-1.2	1.5	0.0
0.9D+1.6WL5	0.6	3.1	0.0
0.9D+1.6WL6	0.7	1.5	0.0
0.9D+1.6WL7	0.6	3.1	0.0
0.9D+1.6WL8	0.8	1.5	0.0

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DESIGN OF FOUNDATION MKD. F1

	About-X	About-Z
Size of column :	0.55m	0.55m
Size of fdn. raft :	1.75m	1.75m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

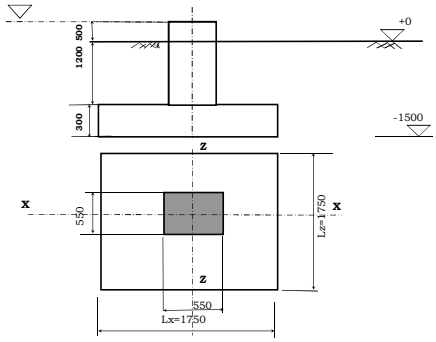
Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W
Crane Load	CL

Calculation of Weight of Foundation :

Weight of raft & Col.= (1.75*1.75*0.3*2.5)+(0.55*0.55*1.7*2.5) = 3.58T

Weight of soil = ((1.75*1.75)-(0.55*0.55))*1.2*1.8 = 5.96T

U/S Base Pl Total = 9.54T



Design Input:

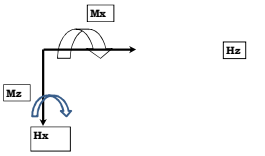
Characteristic strength of concrete, f'$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 217mm = 0.217m

Effective depth of slab along Z direction = 201mm = 0.201m

Clear projection along Z direction = 0.6m

Clear projection along X direction = 0.6m



WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+Lr	2.18	9.27	-0.02	0.0	0.00	18.81	2.18	0.05	-0.02	4.37
DL+CL1	1.26	8.13	-0.01	0.0	0.00	17.67	1.26	0.02	-0.01	2.52
DL+CL2	1.16	8.06	-0.01	0.0	0.00	17.60	1.16	0.02	-0.01	2.32
DL+CL3	1.53	10.91	-0.01	0.0	0.00	20.45	1.53	0.02	-0.01	3.07
DL+CL4	1.19	10.84	-0.01	0.0	0.00	20.38	1.19	0.02	-0.01	2.38
DL+CL5	1.21	8.10	-0.01	0.0	0.00	17.64	1.21	0.02	-0.01	2.42
DL+CL6	1.21	8.10	-0.01	0.0	0.00	17.64	1.21	0.02	-0.01	2.42
DL+CL5	1.36	10.88	-0.01	0.0	0.00	20.42	1.36	0.02	-0.01	2.72
DL+CL6	1.36	10.88	-0.01	0.0	0.00	20.42	1.36	0.02	-0.01	2.72
DL+0.75*CL1+0.75*Lr	2.08	10.00	-0.02	0.0	0.00	19.54	2.08	0.04	-0.02	4.16
DL+0.75*CL2+0.75*Lr	2.00	9.95	-0.02	0.0	0.00	19.49	2.00	0.04	-0.02	4.01
DL+0.75*CL3+0.75*Lr	2.28	12.09	-0.02	0.0	0.00	21.63	2.28	0.04	-0.02	4.56
DL+0.75*CL4+0.75*Lr	2.03	12.04	-0.02	0.0	0.00	21.58	2.03	0.04	-0.02	4.05
DL+0.75*CL5+0.75*Lr	2.04	9.98	-0.02	0.0	0.00	19.52	2.04	0.04	-0.02	4.08
DL+0.75*CL6+0.75*Lr	2.04	9.98	-0.02	0.0	0.00	19.52	2.04	0.04	-0.02	4.08
DL+0.75*CL7+0.75*Lr	2.15	12.06	-0.02	0.0	0.00	21.60	2.15	0.04	-0.02	4.31
DL+0.75*CL8+0.75*Lr	2.15	12.06	-0.02	0.0	0.00	21.60	2.15	0.04	-0.02	4.31
DL+0.75*Lr+0.75*WL1	1.85	7.85	-0.02	0.0	0.00	17.39	1.85	0.03	-0.02	3.71
DL+0.75*Lr+0.75*WL2	1.92	7.14	-0.02	0.0	0.00	16.68	1.92	0.03	-0.02	3.85
DL+0.75*Lr+0.75*WL3	0.83	7.27	-0.02	0.0	0.00	16.81	0.83	0.03	-0.02	1.66
DL+0.75*Lr+0.75*WL4	0.90	6.59	-0.01	0.0	0.00	16.13	0.90	0.03	-0.01	1.79
DL+0.75*Lr+0.75*WL5	1.77	7.37	-0.02	0.0	0.00	16.91	1.77	0.03	-0.02	3.54
DL+0.75*Lr+0.75*WL6	1.79	6.59	-0.01	0.0	0.00	16.13	1.79	0.03	-0.01	3.59
DL+0.75*Lr+0.75*WL7	1.75	7.35	-0.02	0.0	0.00	16.89	1.75	0.03	-0.02	3.50
DL+0.75*Lr+0.75*WL8	1.85	6.59	-0.01	0.0	0.00	16.13	1.85	0.03	-0.01	3.71
0.6*DL+WL1	0.56	2.83	0.00	0.0	0.00	12.37	0.56	0.00	0.00	1.12
0.6*DL+WL2	0.65	1.87	0.00	0.0	0.00	11.41	0.65	0.00	0.00	1.30
0.6*DL+WL3	-0.81	2.05	0.00	0.0	0.00	11.59	-0.81	0.00	0.00	1.62
0.6*DL+WL4	-0.72	1.15	0.00	0.0	0.00	10.69	-0.72	0.01	0.00	1.44
0.6*DL+WL5	0.44	2.18	0.00	0.0	0.00	11.72	0.44	0.00	0.00	0.89
0.6*DL+WL6	0.48	1.14	0.00	0.0	0.00	10.68	0.48	0.00	0.00	0.95
0.6*DL+WL7	0.42	2.16	0.00	0.0	0.00	11.70	0.42	0.00	0.00	0.83
0.6*DL+WL8	0.56	1.15	0.00	0.0	0.00	10.69	0.56	0.00	0.00	1.11

CHECK FOR OVERTURNING AND SLIDING

	Load at Foundation Base					Overturning & Sliding Check (Coeffecient, μ = 0.4)			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
DL+Lr	18.81	2.18	4.37	-0.02	0.05	3.77	351.69	3.45	321.54
DL+CL1	17.67	1.26	2.52	-0.01	0.02	6.14	743.50	5.61	679.77
DL+CL2	17.60	1.16	2.32	-0.01	0.02	6.63	740.44	6.06	676.97
DL+CL3	20.45	1.53	3.07	-0.01	0.02	5.84	860.31	5.34	786.57
DL+CL4	20.38	1.19	2.38	-0.01	0.02	7.50	979.97	6.85	895.97
DL+CL5	17.64	1.21	2.42	-0.01	0.02	6.38	741.97	5.83	678.37
DL+CL6	17.64	1.21	2.42	-0.01	0.02	6.37	659.53	5.82	603.00
DL+CL5	20.42	1.36	2.72	-0.01	0.02	6.56	858.89	6.00	785.27
DL+CL6	20.42	1.36	2.72	-0.01	0.02	6.56	858.89	6.00	785.27
DL+0.75*CL1+0.75*Lr	19.54	2.08	4.16	-0.02	0.04	4.12	424.33	3.76	387.96
DL+0.75*CL2+0.75*Lr	19.49	2.00	4.01	-0.02	0.04	4.25	423.15	3.89	386.88
DL+0.75*CL3+0.75*Lr	21.63	2.28	4.56	-0.02	0.04	4.15	469.55	3.79	429.30
DL+0.75*CL4+0.75*Lr	21.58	2.03	4.05	-0.02	0.04	4.66	492.27	4.26	450.07
DL+0.75*CL5+0.75*Lr	19.52	2.04	4.08	-0.02	0.04	4.18	423.74	3.83	387.42
DL+0.75*CL6+0.75*Lr	19.52	2.04	4.08	-0.02	0.04	4.18	404.18	3.82	369.54
DL+0.75*CL7+0.75*Lr	21.60	2.15	4.31	-0.02	0.04	4.39	469.00	4.01	428.80
DL+0.75*CL8+0.75*Lr	21.60	2.15	4.31	-0.02	0.04	4.39	469.00	4.01	428.80
DL+0.75*Lr+0.75*WL1	17.39	1.85	3.71	-0.02	0.03	4.10	468.29	3.75	428.15
DL+0.75*Lr+0.75*WL2	16.68	1.92	3.85	-0.02	0.03	3.79	477.60	3.47	436.66
DL+0.75*Lr+0.75*WL3	16.81	0.83	1.66	-0.02	0.03	8.88	481.48	8.12	440.21
DL+0.75*Lr+0.75*WL4	16.13	0.90	1.79	-0.01	0.03	7.87	529.73	7.19	484.32
DL+0.75*Lr+0.75*WL5	16.91	1.77	3.54	-0.02	0.03	4.18	455.19	3.82	416.17
DL+0.75*Lr+0.75*WL6	16.13	1.79	3.59	-0.01	0.03	3.94	493.46	3.60	451.16
DL+0.75*Lr+0.75*WL7	16.89	1.75	3.50	-0.02	0.03	4.23	454.74	3.86	415.77
DL+0.75*Lr+0.75*WL8	16.13	1.85	3.71	-0.01	0.03	3.81	493.58	3.48	451.27
0.6*DL+WL1	12.37	0.56	1.12	0.00	0.00	9.70	5202.78	8.87	4756.83
0.6*DL+WL2	11.41	0.65	1.30	0.00	0.00	7.68	19198.95	7.02	17553.32
0.6*DL+WL3	11.59	-0.81	1.62	0.00	0.00	6.25	19503.01	5.72	17831.32
0.6*DL+WL4	10.69	-0.72	1.44	0.00	0.01	6.51	1634.99	5.95	1494.85
0.6*DL+WL5	11.72	0.44	0.89	0.00	0.00	11.56	4929.89	10.57	4507.33
0.6*DL+WL6	10.68	0.48	0.95	0.00	0.00	9.82	2995.66	8.98	2738.89
0.6*DL+WL7	11.70	0.42	0.83	0.00	0.00	12.29	4920.60	11.24	4498.83
0.6*DL+WL8	10.69	0.56	1.11	0.00	0.00	8.42	2997.12	7.70	2740.22

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-05-2020	01-06-2020	25-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	06-05-2020	01-06-2020	25-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

ULTIMATE STRESS DESIGN	Load at U/S Of Base Pit					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4D	1.41	8.54	-0.01	0.0	0.00	18.079	0.03	2.83
1.4D+1.7(CL1)+0.5Lr	2.43	13.58	-0.02	0.0	0.00	23.123	0.04	4.85
1.4D+1.7(CL2)+0.5Lr	2.26	13.46	-0.02	0.0	0.00	23.000	0.04	4.52
1.4D+1.7(CL3)+0.5Lr	2.89	18.30	-0.02	0.0	0.00	27.844	0.04	5.78
1.4D+1.7(CL4)+0.5Lr	2.31	18.19	-0.02	0.0	0.00	27.729	0.04	4.61
1.4D+1.7(CL5)+0.5Lr	2.34	13.52	-0.02	0.0	0.00	23.062	0.04	4.68
1.4D+1.7(CL6)+0.5Lr	2.34	13.52	-0.02	0.0	0.00	23.062	0.05	4.69
1.4D+1.7(CL5)+0.5Lr	2.60	18.25	-0.02	0.0	0.00	27.787	0.04	5.20
1.4D+1.7(CL6)+0.5Lr	2.60	18.25	-0.02	0.0	0.00	27.787	0.04	5.20
1.2D+1.6Lr+1.0(CL1)	3.34	14.43	-0.03	0.0	0.00	23.967	0.07	6.68
1.2D+1.6Lr+1.0(CL2)	3.24	14.35	-0.03	0.0	0.00	23.894	0.07	6.48
1.2D+1.6Lr+1.0(CL3)	3.61	17.20	-0.03	0.0	0.00	26.744	0.07	7.23
1.2D+1.6Lr+1.0(CL4)	3.27	17.14	-0.03	0.0	0.00	26.676	0.06	6.54
1.2D+1.6Lr+1.0(CL5)	3.29	14.39	-0.03	0.0	0.00	23.931	0.07	6.58
1.2D+1.6Lr+1.0(CL6)	3.29	14.39	-0.03	0.0	0.00	23.931	0.07	6.58
1.2D+1.6Lr+1.0(CL7)	3.44	17.17	-0.03	0.0	0.00	26.710	0.07	6.88
1.2D+1.6Lr+1.0(CL8)	3.44	17.17	-0.03	0.0	0.00	26.710	0.07	6.88
1.2D + 1.6WL1+1.0L+0.5Lr	1.72	7.57	-0.01	0.0	0.00	17.114	0.02	3.44
1.2D + 1.6WL2+1.0L+0.5Lr	1.87	6.04	-0.01	0.0	0.00	15.581	0.02	3.74
1.2D + 1.6WL3+1.0L+0.5Lr	-0.47	6.33	-0.01	0.0	0.00	15.870	0.02	0.94
1.2D + 1.6WL4+1.0L+0.5Lr	-0.32	4.89	0.00	0.0	0.00	14.426	0.01	0.64
1.2D + 1.6WL5+1.0L+0.5Lr	1.54	6.54	-0.01	0.0	0.00	16.076	0.02	3.08
1.2D + 1.6WL6+1.0L+0.5Lr	1.59	4.88	-0.01	0.0	0.00	14.416	0.01	3.18
1.2D + 1.6WL7+1.0L+0.5Lr	1.50	6.50	-0.01	0.0	0.00	16.040	0.02	2.99
1.2D + 1.6WL8+1.0L+0.5Lr	1.72	4.88	-0.01	0.0	0.00	14.424	0.01	3.44
0.9D+1.6WL1	0.83	4.16	0.00	0.0	0.00	13.698	0.00	1.66
0.9D+1.6WL2	0.98	2.63	0.00	0.0	0.00	12.165	0.00	1.96
0.9D+1.6WL3	-1.36	2.91	0.00	0.0	0.00	12.455	0.00	2.72
0.9D+1.6WL4	-1.21	1.47	0.01	0.0	0.00	11.011	0.01	2.42
0.9D+1.6WL5	0.65	3.12	0.00	0.0	0.00	12.661	0.00	1.30
0.9D+1.6WL6	0.70	1.46	0.00	0.0	0.00	11.001	0.01	1.40
0.9D+1.6WL7	0.61	3.09	0.00	0.0	0.00	12.625	0.00	1.21
0.9D+1.6WL8	0.83	1.47	0.00	0.0	0.00	11.009	0.01	1.66

Check for Bearing Pressure

Load Combination & Calculation of Base pressure

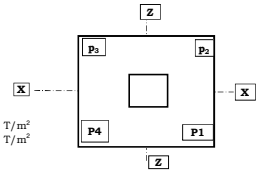
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 (BH28) For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 53.6 T/m^2 For WL/SL case = 66.1 T/m^2



Load Case	Design Load/moments			Base Pressure (T/m^2)								
	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-p_h)	Remarks
DL+Lr	18.81	4.37	0.05	6.14	4.89	0.05	13.08	12.98	3.30	3.20	10.92	O.K
DL+CL1	17.67	2.52	0.02	5.77	2.82	0.02	10.61	10.57	4.97	4.93	8.45	O.K
DL+CL2	17.60	2.32	0.02	5.75	2.60	0.02	10.37	10.33	5.17	5.12	8.21	O.K
DL+CL3	20.45	3.07	0.02	6.68	3.43	0.02	12.13	12.09	5.27	5.22	9.97	O.K
DL+CL4	20.38	2.38	0.02	6.66	2.66	0.02	11.34	11.30	6.01	5.97	9.18	O.K
DL+CL5	17.64	2.42	0.02	5.76	2.71	0.02	10.49	10.45	5.07	5.03	8.33	O.K
DL+CL6	17.64	2.42	0.02	5.76	2.71	0.03	10.50	10.45	5.07	5.02	8.34	O.K
DL+CL5	20.42	2.72	0.02	6.67	3.05	0.02	11.74	11.69	5.64	5.60	9.58	O.K
DL+CL6	20.42	2.72	0.02	6.67	3.05	0.02	11.74	11.69	5.64	5.60	9.58	O.K
DL+0.75*CL1+0.75*Lr	19.54	4.16	0.04	6.38	4.65	0.05	13.08	12.99	3.77	3.68	10.92	O.K
DL+0.75*CL2+0.75*Lr	19.49	4.01	0.04	6.36	4.49	0.05	12.90	12.81	3.92	3.83	10.74	O.K
DL+0.75*CL3+0.75*Lr	21.63	4.56	0.04	7.06	5.11	0.05	14.22	14.13	4.00	3.91	12.06	O.K
DL+0.75*CL4+0.75*Lr	21.58	4.05	0.04	7.05	4.53	0.04	13.62	13.54	4.55	4.47	11.46	O.K
DL+0.75*CL5+0.75*Lr	19.52	4.08	0.04	6.37	4.57	0.05	12.99	12.90	3.85	3.76	10.83	O.K
DL+0.75*CL6+0.75*Lr	19.52	4.08	0.04	6.37	4.57	0.05	12.99	12.90	3.85	3.75	10.83	O.K
DL+0.75*CL7+0.75*Lr	21.60	4.31	0.04	7.05	4.82	0.05	13.92	13.83	4.28	4.19	11.76	O.K
DL+0.75*CL8+0.75*Lr	21.60	4.31	0.04	7.05	4.82	0.05	13.92	13.83	4.28	4.19	11.76	O.K
DL+0.75*Lr+0.75*WL1	17.39	3.71	0.03	5.68	4.15	0.04	11.87	11.80	3.56	3.49	9.71	O.K
DL+0.75*Lr+0.75*WL2	16.68	3.85	0.03	5.44	4.31	0.03	11.79	11.72	3.17	3.10	9.63	O.K
DL+0.75*Lr+0.75*WL3	16.81	1.66	0.03	5.49	1.85	0.03	9.38	9.31	5.67	5.60	7.22	O.K
DL+0.75*Lr+0.75*WL4	16.13	1.79	0.03	5.27	2.01	0.03	9.31	9.25	5.29	5.23	7.15	O.K
DL+0.75*Lr+0.75*WL5	16.91	3.54	0.03	5.52	3.96	0.04	11.52	11.44	3.60	3.52	9.36	O.K
DL+0.75*Lr+0.75*WL6	16.13	3.59	0.03	5.27	4.01	0.03	11.31	11.25	3.28	3.22	9.15	O.K
DL+0.75*Lr+0.75*WL7	16.89	3.50	0.03	5.52	3.91	0.04	11.47	11.39	3.64	3.56	9.31	O.K
DL+0.75*Lr+0.75*WL8	16.13	3.71	0.03	5.27	4.15	0.03	11.45	11.38	3.15	3.09	9.29	O.K
0.6*DL+WL1	12.37	1.12	0.00	4.04	1.25	0.00	7.29	7.29	4.79	4.79	5.13	O.K
0.6*DL+WL2	11.41	1.30	0.00	3.73	1.46	0.00	7.18	7.18	4.27	4.27	5.02	O.K
0.6*DL+WL3	11.59	1.62	0.00	3.78	1.82	0.00	7.60	7.60	3.97	3.97	5.44	O.K
0.6*DL+WL4	10.69	1.44	0.01	3.49	1.61	0.01	7.11	7.09	3.89	3.87	4.95	O.K
0.6*DL+WL5	11.72	0.89	0.00	3.83	0.99	0.00	6.82	6.82	4.84	4.83	4.66	O.K
0.6*DL+WL6	10.68	0.95	0.00	3.49	1.07	0.00	6.56	6.55	4.43	4.42	4.40	O.K
0.6*DL+WL7	11.70	0.83	0.00	3.82	0.93	0.00	6.75	6.75	4.89	4.89	4.59	O.K
0.6*DL+WL8	10.69	1.11	0.00	3.49	1.24	0.00	6.74	6.73	4.25	4.24	4.58	O.K

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-05-2020	01-06-2020	25-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	06-05-2020	01-06-2020	25-10-2023	
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Prepared By	SHD	SHD	SHD	
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Approved	PRI	PRI	PRI	

Calculation of Design Pressure

Load Combination & Calculation of Base pressure

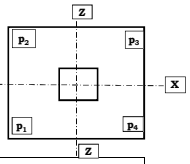
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 — For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 53.6 T/m^2 — For WL/SL case = 66.1 T/m^2



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	P_1 (p_{max})	P_2	P_4	P_3 (p_{min})	(p_{max}-p_3)
1.4D	18.1	2.83	0.03	5.90	3.17	0.03	11.10	11.04	4.77	4.70	8.40
1.4D+1.7(CL1)+0.5Lr	23.1	4.85	0.04	7.55	5.43	0.05	15.03	12.93	2.17	2.07	12.33
1.4D+1.7(CL2)+0.5Lr	23.0	4.52	0.04	7.51	5.06	0.05	14.62	12.52	2.50	2.40	11.92
1.4D+1.7(CL3)+0.5Lr	27.8	5.78	0.04	9.09	6.47	0.05	17.61	15.51	2.67	2.57	14.91
1.4D+1.7(CL4)+0.5Lr	27.7	4.61	0.04	9.05	5.16	0.04	16.26	14.18	3.93	3.85	13.56
1.4D+1.7(CL5)+0.5Lr	23.1	4.68	0.04	7.53	5.24	0.05	14.82	12.73	2.33	2.24	12.12
1.4D+1.7(CL6)+0.5Lr	23.1	4.69	0.05	7.53	5.25	0.05	14.83	12.73	2.33	2.23	12.13
1.4D+1.7(CL5)+0.5Lr	27.8	5.20	0.04	9.07	5.82	0.05	16.94	14.84	3.30	3.21	14.24
1.4D+1.7(CL6)+0.5Lr	27.8	5.20	0.04	9.07	5.82	0.05	16.94	14.84	3.30	3.21	14.24
1.2D+1.6Lr+1.0(CL1)	24.0	6.68	0.07	7.83	7.48	0.07	17.38	15.23	0.42	0.27	14.68
1.2D+1.6Lr+1.0(CL2)	23.9	6.48	0.07	7.80	7.26	0.07	17.14	14.99	0.62	0.47	14.44
1.2D+1.6Lr+1.0(CL3)	26.7	7.23	0.07	8.73	8.09	0.07	18.90	16.75	0.72	0.57	16.20
1.2D+1.6Lr+1.0(CL4)	26.7	6.54	0.06	8.71	7.32	0.07	18.10	15.96	1.46	1.32	15.40
1.2D+1.6Lr+1.0(CL5)	23.9	6.58	0.07	7.81	7.37	0.07	17.26	15.11	0.52	0.37	14.56
1.2D+1.6Lr+1.0(CL6)	23.9	6.58	0.07	7.81	7.37	0.08	17.26	15.11	0.52	0.37	14.56
1.2D+1.6Lr+1.0(CL7)	26.7	6.88	0.07	8.72	7.71	0.07	18.50	16.35	1.09	0.94	15.80
1.2D+1.6Lr+1.0(CL8)	26.7	6.88	0.07	8.72	7.71	0.07	18.50	16.35	1.09	0.94	15.80
1.2D + 1.6WL1+1.0L+0.5Lr	17.1	3.44	0.02	5.59	3.86	0.02	11.47	9.42	1.76	1.71	8.77
1.2D + 1.6WL2+1.0L+0.5Lr	15.6	3.74	0.02	5.09	4.19	0.02	11.29	9.25	0.92	0.88	8.59
1.2D + 1.6WL3+1.0L+0.5Lr	15.9	0.94	0.02	5.18	1.05	0.02	8.25	6.21	4.15	4.11	5.55
1.2D + 1.6WL4+1.0L+0.5Lr	14.4	0.64	0.01	4.71	0.72	0.01	7.44	5.42	4.00	3.98	4.74
1.2D + 1.6WL5+1.0L+0.5Lr	16.1	3.08	0.02	5.25	3.45	0.02	10.72	8.67	1.83	1.78	8.02
1.2D + 1.6WL6+1.0L+0.5Lr	14.4	3.18	0.01	4.71	3.56	0.01	10.28	8.26	1.16	1.13	7.58
1.2D + 1.6WL7+1.0L+0.5Lr	16.0	2.99	0.02	5.24	3.35	0.02	10.61	8.56	1.91	1.87	7.91
1.2D + 1.6WL8+1.0L+0.5Lr	14.4	3.44	0.01	4.71	3.85	0.01	10.57	8.54	0.88	0.85	7.87
0.9D+1.6WL1	13.7	1.66	0.00	4.47	1.86	0.00	8.34	6.33	2.61	2.61	5.64
0.9D+1.6WL2	12.2	1.96	0.00	3.97	2.19	0.00	8.17	6.16	1.78	1.78	5.47
0.9D+1.6WL3	12.5	2.72	0.00	4.07	3.04	0.00	9.11	7.11	1.03	1.02	6.41
0.9D+1.6WL4	11.0	2.42	0.01	3.60	2.71	0.01	8.32	6.29	0.90	0.87	5.62
0.9D+1.6WL5	12.7	1.30	0.00	4.13	1.45	0.00	7.59	5.59	2.68	2.68	4.89
0.9D+1.6WL6	11.0	1.40	0.01	3.59	1.57	0.01	7.17	5.15	2.03	2.02	4.47
0.9D+1.6WL7	12.6	1.21	0.00	4.12	1.36	0.00	7.48	5.48	2.77	2.76	4.78
0.9D+1.6WL8	11.0	1.66	0.01	3.59	1.85	0.01	7.46	5.44	1.75	1.73	4.76

Global Project ID	900707
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Ma'aden Revision No:	A01	C01	D01
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Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

SF (One way)= Oneway shear force at a distance 'd' from the face of column in T/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.55+2*0.5*0.201)+(0.55+2*0.5*0.201))*2

= 3.004 m

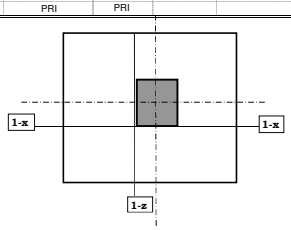
Punching Area= ((0.55+2*0.5*0.201)*(0.55+2*0.5*0.201))

= 0.56 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.3*2.5+1.2*1.8)

= 2.91 T/m^2

Area of footing, A_f= 3.06 m^2



Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 40.54 T

Strength of Concrete in shear Vc1 = 129.14 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 139.21 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 119.07 T **Table 22.6.5.2_ (c)**

Design Vc= 119.07 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (40.54*10000)/(3004*201)

= 0.67 N/mm^2

No shear reinforcement reqd.

Load Case	p_{1x} (T/m^2)	$B.M_{1x}$ T-m	$S.F_{(oneway)}$ (T)	$B.M_{1z}$	$S.F_{(Oneway)}$ (T)	$S.F_{(punching)}$	V_u/b_0d	Punching check (Remarks)	Allowable shear (0.75*√(fc)/6 (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4D	8.40	1.51	3.22	1.51	3.35	21.02	0.35	OK	0.74	0.167	OK	0.37
1.4D+1.7(CL1)+0.5Lr	12.33	2.22	4.72	2.22	4.92	30.86	0.51	OK	0.74	0.245	OK	0.55
1.4D+1.7(CL2)+0.5Lr	11.92	2.15	4.57	2.15	4.76	29.83	0.49	OK	0.74	0.237	OK	0.53
1.4D+1.7(CL3)+0.5Lr	14.91	2.68	5.71	2.68	5.95	37.31	0.62	OK	0.74	0.296	OK	0.66
1.4D+1.7(CL4)+0.5Lr	13.56	2.44	5.19	2.44	5.41	33.93	0.56	OK	0.74	0.269	OK	0.60
1.4D+1.7(CL5)+0.5Lr	12.12	2.18	4.64	2.18	4.84	30.33	0.5	OK	0.74	0.241	OK	0.54
1.4D+1.7(CL6)+0.5Lr	12.13	2.18	4.65	2.18	4.84	30.36	0.5	OK	0.74	0.241	OK	0.54
1.4D+1.7(CL5)+0.5Lr	14.24	2.56	5.45	2.56	5.68	35.64	0.59	OK	0.74	0.283	OK	0.63
1.4D+1.7(CL6)+0.5Lr	14.24	2.56	5.45	2.56	5.68	35.64	0.59	OK	0.74	0.283	OK	0.63
1.2D+1.6Lr+1.0(CL1)	14.68	2.64	5.62	2.64	5.86	36.74	0.61	OK	0.74	0.291	OK	0.65
1.2D+1.6Lr+1.0(CL2)	14.44	2.60	5.53	2.60	5.76	36.14	0.6	OK	0.74	0.287	OK	0.64
1.2D+1.6Lr+1.0(CL3)	16.20	2.92	6.20	2.92	6.46	40.54	0.67	OK	0.74	0.322	OK	0.72
1.2D+1.6Lr+1.0(CL4)	15.40	2.77	5.90	2.77	6.14	38.54	0.64	OK	0.74	0.306	OK	0.69
1.2D+1.6Lr+1.0(CL5)	14.56	2.62	5.58	2.62	5.81	36.44	0.6	OK	0.74	0.289	OK	0.65
1.2D+1.6Lr+1.0(CL6)	14.56	2.62	5.58	2.62	5.81	36.44	0.6	OK	0.74	0.289	OK	0.65
1.2D+1.6Lr+1.0(CL7)	15.80	2.84	6.05	2.84	6.30	39.54	0.65	OK	0.74	0.314	OK	0.70
1.2D+1.6Lr+1.0(CL8)	15.80	2.84	6.05	2.84	6.30	39.54	0.65	OK	0.74	0.314	OK	0.70
1.2D + 1.6WL1+1.0L+0.5Lr	8.77	1.58	3.36	1.58	3.50	21.95	0.36	OK	0.74	0.174	OK	0.39
1.2D + 1.6WL2+1.0L+0.5Lr	8.59	1.55	3.29	1.55	3.43	21.50	0.36	OK	0.74	0.171	OK	0.38
1.2D + 1.6WL3+1.0L+0.5Lr	5.55	1.00	2.13	1.00	2.21	13.89	0.23	OK	0.74	0.11	OK	0.25
1.2D + 1.6WL4+1.0L+0.5Lr	4.74	0.85	1.82	0.85	1.89	11.86	0.2	OK	0.74	0.094	OK	0.21
1.2D + 1.6WL5+1.0L+0.5Lr	8.02	1.44	3.07	1.44	3.20	20.07	0.33	OK	0.74	0.159	OK	0.36
1.2D + 1.6WL6+1.0L+0.5Lr	7.58	1.36	2.90	1.36	3.02	18.97	0.31	OK	0.74	0.15	OK	0.34
1.2D + 1.6WL7+1.0L+0.5Lr	7.91	1.42	3.03	1.42	3.16	19.79	0.33	OK	0.74	0.157	OK	0.35
1.2D + 1.6WL8+1.0L+0.5Lr	7.87	1.42	3.01	1.42	3.14	19.69	0.33	OK	0.74	0.156	OK	0.35
0.9D+1.6WL1	5.64	1.02	2.16	1.02	2.25	14.11	0.23	OK	0.74	0.112	OK	0.25
0.9D+1.6WL2	5.47	0.98	2.10	0.98	2.18	13.69	0.23	OK	0.74	0.109	OK	0.24
0.9D+1.6WL3	6.41	1.15	2.46	1.15	2.56	16.04	0.27	OK	0.74	0.127	OK	0.29
0.9D+1.6WL4	5.62	1.01	2.15	1.01	2.24	14.06	0.23	OK	0.74	0.112	OK	0.25
0.9D+1.6WL5	4.89	0.88	1.87	0.88	1.95	12.24	0.2	OK	0.74	0.097	OK	0.22
0.9D+1.6WL6	4.47	0.80	1.71	0.80	1.78	11.19	0.19	OK	0.74	0.089	OK	0.20
0.9D+1.6WL7	4.78	0.86	1.83	0.86	1.91	11.96	0.2	OK	0.74	0.095	OK	0.21
0.9D+1.6WL8	4.76	0.86	1.82	0.86	1.90	11.91	0.2	OK	0.74	0.094	OK	0.21

Load Case	kn (Ø=0.75)	p =	A_{st}
Ultimate Conditions			
1.4D	0.5	0.001	217
1.4D+1.7(CL1)+0.5Lr	0.73	0.002	434
1.4D+1.7(CL2)+0.5Lr	0.71	0.002	434
1.4D+1.7(CL3)+0.5Lr	0.89	0.002	434
1.4D+1.7(CL4)+0.5Lr	0.81	0.002	434
1.4D+1.7(CL5)+0.5Lr	0.72	0.002	434
1.4D+1.7(CL6)+0.5Lr	0.72	0.002	434
1.4D+1.7(CL5)+0.5Lr	0.85	0.002	434
1.4D+1.7(CL6)+0.5Lr	0.85	0.002	434
1.2D+1.6Lr+1.0(CL1)	0.87	0.002	434
1.2D+1.6Lr+1.0(CL2)	0.86	0.002	434
1.2D+1.6Lr+1.0(CL3)	0.96	0.002	434
1.2D+1.6Lr+1.0(CL4)	0.91	0.002	434
1.2D+1.6Lr+1.0(CL5)	0.86	0.002	434
1.2D+1.6Lr+1.0(CL6)	0.86	0.002	434
1.2D+1.6Lr+1.0(CL7)	0.94	0.002	434
1.2D+1.6Lr+1.0(CL8)	0.94	0.002	434
1.2D + 1.6WL1+1.0L+0.5Lr	0.52	0.001	217
1.2D + 1.6WL2+1.0L+0.5Lr	0.51	0.001	217
1.2D + 1.6WL3+1.0L+0.5Lr	0.33	0.001	217
1.2D + 1.6WL4+1.0L+0.5Lr	0.28	0.001	217
1.2D + 1.6WL5+1.0L+0.5Lr	0.48	0.001	217
1.2D + 1.6WL6+1.0L+0.5Lr	0.45	0.001	217
1.2D + 1.6WL7+1.0L+0.5Lr	0.47	0.001	217
1.2D + 1.6WL8+1.0L+0.5Lr	0.47	0.001	217
0.9D+1.6WL1	0.34	0.001	217
0.9D+1.6WL2	0.32	0.001	217
0.9D+1.6WL3	0.38	0.001	217
0.9D+1.6WL4	0.33	0.001	217
0.9D+1.6WL5	0.29	0.001	217
0.9D+1.6WL6	0.27	0.001	217
0.9D+1.6WL7	0.28	0.001	217
0.9D+1.6WL8	0.28	0.001	217

Location	Mux (T-m)	ϕ Kn = M_{ux}/bd^2	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing	Spacing provd.
At bottom	2.92	0.72	0.96	0.002	434.00	390.60	12øT	260.59	300	200
At top	0.00	0.00	0.00	0.0018	361.80	361.80	12øT	312.60	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**12øT**	**200**	c/c	At Top	**12øT**	**200**	c/c
Along Z Direction :	At Bottom	**12øT**	**200**	c/c	At Top	**12øT**	**200**	c/c

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2	3		
Date	06-05-2020	01-06-2020	25-10-2023				Date	06-05-2020	01-06-2020	25-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

3.2 Design of Foundation F2 (GRID A- 1 , 4, 5 6)

	A 1,4,5,6		
LOAD DEFINATION	Fx Ton	Fy Ton	Fz Ton
1 DL	-1.1	6.2	0.0
2 LLROOF	-1.2	3.2	0.0
3 CL1(L+XS)	-0.2	4.7	0.0
4 CL2(L-XS)	-0.5	4.8	0.0
5 CL3(R+XS)	-0.2	2.0	0.0
6 CL4(R-XS)	-0.3	2.0	0.0
7 CL5(L+ZT)	-0.3	4.8	0.0
8 CL6(L-ZT)	-0.3	4.8	0.0
9 CL7(R+ZT)	-0.2	2.0	0.0
10 CL8(R-ZT)	-0.2	2.0	0.0
21 WL1(+X)S	1.5	-1.7	0.0
22 WL2(+X)P	1.4	-2.6	0.0
23 WL3(-X)S	0.1	-0.9	0.0
24 WL4(-X)P	0.0	-1.6	0.0
25 WL5(+Z)S	0.2	-1.5	0.0
26 WL6(+Z)P	0.2	-2.6	0.0
27 WL7(-Z)S	0.2	-1.5	0.0
28 WL8(-Z)P	0.1	-2.6	0.0

STRENGTH DESIGN	GRID A 1,4 , 5 ,6		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4D	-1.48	8.63	-0.01
1.4D+1.7(CL1)+0.5Lr	-2.39	18.30	-0.02
1.4D+1.7(CL2)+0.5Lr	-2.98	18.42	-0.02
1.4D+1.7(CL3)+0.5Lr	-2.35	13.58	-0.02
1.4D+1.7(CL4)+0.5Lr	-2.52	13.70	-0.02
1.4D+1.7(CL5)+0.5Lr	-2.69	18.36	-0.02
1.4D+1.7(CL6)+0.5Lr	-2.69	18.36	-0.02
1.4D+1.7(CL7)+0.5Lr	-2.44	13.64	-0.02
1.4D+1.7(CL8)+0.5Lr	-2.44	13.64	-0.02
1.2D+1.6Lr+1.0(CL1)	-3.42	17.31	-0.03
1.2D+1.6Lr+1.0(CL2)	-3.77	17.38	-0.03
1.2D+1.6Lr+1.0(CL3)	-3.40	14.53	-0.03
1.2D+1.6Lr+1.0(CL4)	-3.50	14.60	-0.03
1.2D+1.6Lr+1.0(CL5)	-3.60	17.34	-0.03
1.2D+1.6Lr+1.0(CL6)	-3.60	17.34	-0.03
1.2D+1.6Lr+1.0(CL7)	-3.45	14.56	-0.03
1.2D+1.6Lr+1.0(CL8)	-3.45	14.56	-0.03
1.2D + 1.6WL1+1.0L+0.5Lr	0.50	6.36	-0.01
1.2D + 1.6WL2+1.0L+0.5Lr	0.32	4.92	0.00
1.2D + 1.6WL3+1.0L+0.5Lr	-1.74	7.61	-0.01
1.2D + 1.6WL4+1.0L+0.5Lr	-1.92	6.51	-0.01
1.2D + 1.6WL5+1.0L+0.5Lr	-1.58	6.60	-0.01
1.2D + 1.6WL6+1.0L+0.5Lr	-1.60	4.91	-0.01
1.2D + 1.6WL7+1.0L+0.5Lr	-1.64	6.66	-0.01
1.2D + 1.6WL8+1.0L+0.5Lr	-1.73	4.92	-0.01
0.9D+1.6WL1	1.44	2.90	0.00
0.9D+1.6WL2	1.25	1.45	0.01
0.9D+1.6WL3	-0.81	4.15	0.00
0.9D+1.6WL4	-0.98	3.05	0.00
0.9D+1.6WL5	-0.65	3.14	0.00
0.9D+1.6WL6	-0.67	1.44	0.00
0.9D+1.6WL7	-0.71	3.20	0.00
0.9D+1.6WL8	-0.80	1.46	0.00

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Ma'aden Revision No:	A01	C01	D01
Date	06-05-2020	01-06-2020	25-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
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Outotec / L&T Revision No.:	0	1	2	3
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Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

DESIGN OF FOUNDATION MKD.F2

	About-X	About-Z
Size of column :	0.55m	0.55m
Size of fdn. raft :	1.75m	1.75m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

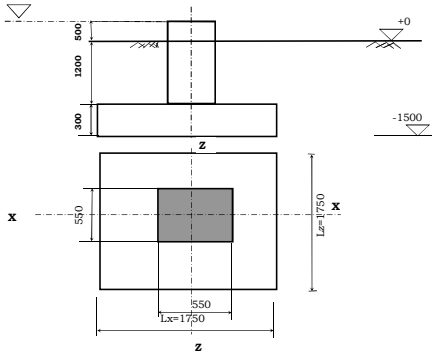
Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W
Crane Load	CL

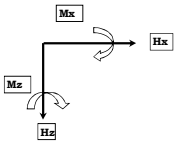
Calculation of Weight of Foundation :

Weight of raft & Col.= (1.75*1.75*0.3*2.5)+(0.55*0.55*1.7*2.5) = 3.58T

Weight of soil = ((1.75*1.75)-(0.55*0.55))*1.2*1.8 = 5.96T

U/S Base Pl Total = 9.54T





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	20mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	20mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 215mm = 0.215m

Effective depth of slab along Z direction = 195mm = 0.195m

Clear projection along Z direction = 0.6m

Clear projection along X direction = 0.6m

WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+Lr	-2.29	9.39	-0.02	0.0	0.00	18.93	-2.29	0.05	-0.02	4.59
DL+CL1	-1.23	10.90	-0.01	0.0	0.00	20.44	-1.23	0.02	-0.01	2.46
DL+CL2	-1.58	10.97	-0.01	0.0	0.00	20.51	-1.58	0.02	-0.01	3.16
DL+CL3	-1.21	8.12	-0.01	0.0	0.00	17.66	-1.21	0.02	-0.01	2.42
DL+CL4	-1.31	8.20	-0.01	0.0	0.00	17.74	-1.31	0.02	-0.01	2.62
DL+CL5	-1.41	10.94	-0.01	0.0	0.00	20.48	-1.41	0.02	-0.01	2.81
DL+CL6	-1.41	10.94	-0.01	0.0	0.00	20.48	-1.41	0.02	-0.01	2.81
DL+CL5	-1.26	8.16	-0.01	0.0	0.00	17.70	-1.26	0.02	-0.01	2.51
DL+CL6	-1.26	8.16	-0.01	0.0	0.00	17.70	-1.26	0.02	-0.01	2.51
DL+0.75*CL1+0.75*Lr	-2.12	12.14	-0.02	0.0	0.00	21.68	-2.12	0.04	-0.02	4.23
DL+0.75*CL2+0.75*Lr	-2.38	12.19	-0.02	0.0	0.00	21.73	-2.38	0.04	-0.02	4.75
DL+0.75*CL3+0.75*Lr	-2.10	10.06	-0.02	0.0	0.00	19.60	-2.10	0.04	-0.02	4.20
DL+0.75*CL4+0.75*Lr	-2.17	10.11	-0.02	0.0	0.00	19.65	-2.17	0.04	-0.02	4.35
DL+0.75*CL5+0.75*Lr	-2.25	12.17	-0.02	0.0	0.00	21.71	-2.25	0.04	-0.02	4.49
DL+0.75*CL6+0.75*Lr	-2.25	12.17	-0.02	0.0	0.00	21.71	-2.25	0.04	-0.02	4.49
DL+0.75*CL7+0.75*Lr	-2.14	10.08	-0.02	0.0	0.00	19.62	-2.14	0.04	-0.02	4.27
DL+0.75*CL8+0.75*Lr	-2.14	10.08	-0.02	0.0	0.00	19.62	-2.14	0.04	-0.02	4.27
DL+0.75*Lr+0.75*WL1	-0.86	7.35	-0.02	0.0	0.00	16.89	-0.86	0.03	-0.02	1.73
DL+0.75*Lr+0.75*WL2	-0.95	6.67	-0.01	0.0	0.00	16.21	-0.95	0.03	-0.01	1.91
DL+0.75*Lr+0.75*WL3	-1.92	7.93	-0.02	0.0	0.00	17.47	-1.92	0.03	-0.02	3.84
DL+0.75*Lr+0.75*WL4	-2.00	7.42	-0.01	0.0	0.00	16.96	-2.00	0.03	-0.01	4.00
DL+0.75*Lr+0.75*WL5	-1.84	7.46	-0.02	0.0	0.00	17.00	-1.84	0.03	-0.02	3.68
DL+0.75*Lr+0.75*WL6	-1.85	6.66	-0.01	0.0	0.00	16.20	-1.85	0.03	-0.01	3.71
DL+0.75*Lr+0.75*WL7	-1.87	7.49	-0.02	0.0	0.00	17.03	-1.87	0.03	-0.02	3.74
DL+0.75*Lr+0.75*WL8	-1.91	6.67	-0.01	0.0	0.00	16.21	-1.91	0.03	-0.01	3.83
0.6*DL+WL1	0.86	2.04	0.00	0.0	0.00	11.58	0.86	0.00	0.00	1.72
0.6*DL+WL2	0.74	1.14	0.00	0.0	0.00	10.68	0.74	0.01	0.00	1.49
0.6*DL+WL3	-0.54	2.82	0.00	0.0	0.00	12.36	-0.54	0.00	0.00	1.09
0.6*DL+WL4	-0.65	2.14	0.00	0.0	0.00	11.68	-0.65	0.00	0.00	1.31
0.6*DL+WL5	-0.44	2.19	0.00	0.0	0.00	11.73	-0.44	0.00	0.00	0.89
0.6*DL+WL6	-0.46	1.13	0.00	0.0	0.00	10.67	-0.46	0.00	0.00	0.91
0.6*DL+WL7	-0.48	2.23	0.00	0.0	0.00	11.77	-0.48	0.00	0.00	0.96
0.6*DL+WL8	-0.54	1.14	0.00	0.0	0.00	10.68	-0.54	0.00	0.00	1.08

Check for overturning and sliding:

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
DL+Lr	18.93	-2.29	4.59	-0.02	0.05	3.61	354.00	3.30	323.65
DL+CL1	20.44	-1.23	2.46	-0.01	0.02	7.26	859.99	6.64	786.27
DL+CL2	20.51	-1.58	3.16	-0.01	0.02	5.69	862.88	5.20	788.92
DL+CL3	17.66	-1.21	2.42	-0.01	0.02	6.40	743.06	5.85	679.37
DL+CL4	17.74	-1.31	2.62	-0.01	0.02	5.93	746.07	5.42	682.12
DL+CL5	20.48	-1.41	2.81	-0.01	0.02	6.37	861.41	5.83	787.57
DL+CL6	20.48	-1.41	2.81	-0.01	0.02	6.38	861.46	5.83	787.62
DL+CL5	17.70	-1.26	2.51	-0.01	0.02	6.16	744.54	5.63	680.72
DL+CL6	17.70	-1.26	2.51	-0.01	0.02	6.16	661.81	5.63	605.09
DL+0.75*CL1+0.75*Lr	21.68	-2.12	4.23	-0.02	0.04	4.48	470.76	4.10	430.41
DL+0.75*CL2+0.75*Lr	21.73	-2.38	4.75	-0.02	0.04	4.00	471.88	3.66	431.43
DL+0.75*CL3+0.75*Lr	19.60	-2.10	4.20	-0.02	0.04	4.09	425.50	3.74	389.02
DL+0.75*CL4+0.75*Lr	19.65	-2.17	4.35	-0.02	0.04	3.96	426.66	3.62	390.09
DL+0.75*CL5+0.75*Lr	21.71	-2.25	4.49	-0.02	0.04	4.23	471.31	3.87	430.91
DL+0.75*CL6+0.75*Lr	21.71	-2.25	4.49	-0.02	0.04	4.23	471.33	3.87	430.93
DL+0.75*CL7+0.75*Lr	19.62	-2.14	4.27	-0.02	0.04	4.02	426.07	3.68	389.55
DL+0.75*CL8+0.75*Lr	19.62	-2.14	4.27	-0.02	0.04	4.02	406.40	3.68	371.57
DL+0.75*Lr+0.75*WL1	16.89	-0.86	1.73	-0.02	0.03	8.54	483.63	7.81	442.18
DL+0.75*Lr+0.75*WL2	16.21	-0.95	1.91	-0.01	0.03	7.44	532.16	6.81	486.54
DL+0.75*Lr+0.75*WL3	17.47	-1.92	3.84	-0.02	0.03	3.99	470.34	3.64	430.02
DL+0.75*Lr+0.75*WL4	16.96	-2.00	4.00	-0.01	0.03	3.71	518.75	3.39	474.29
DL+0.75*Lr+0.75*WL5	17.00	-1.84	3.68	-0.02	0.03	4.04	457.58	3.69	418.36
DL+0.75*Lr+0.75*WL6	16.20	-1.85	3.71	-0.01	0.03	3.83	495.73	3.50	453.23
DL+0.75*Lr+0.75*WL7	17.03	-1.87	3.74	-0.02	0.03	3.98	458.37	3.64	419.08
DL+0.75*Lr+0.75*WL8	16.21	-1.91	3.83	-0.01	0.03	3.71	495.90	3.39	453.40
0.6*DL+WL1	11.58	0.86	1.72	0.00	0.00	5.90	19491.63	5.39	17820.92
0.6*DL+WL2	10.68	0.74	1.49	0.00	0.01	6.29	1633.76	5.75	1493.72
0.6*DL+WL3	12.36	-0.54	1.09	0.00	0.00	9.93	5200.49	9.08	4754.73
0.6*DL+WL4	11.68	-0.65	1.31	0.00	0.00	7.80	3274.86	7.13	2994.15
0.6*DL+WL5	11.73	-0.44	0.89	0.00	0.00	11.58	4934.71	10.59	4511.73
0.6*DL+WL6	10.67	-0.46	0.91	0.00	0.00	10.21	2993.40	9.34	2736.82
0.6*DL+WL7	11.77	-0.48	0.96	0.00	0.00	10.71	4951.11	9.79	4526.73
0.6*DL+WL8	10.68	-0.54	1.08	0.00	0.00	8.69	2995.58	7.94	2738.82

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-05-2020	01-06-2020	25-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	06-05-2020	01-06-2020	25-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

ULTIMATE STRESS DESIGN	Load at U/S Of Base Pit					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4D	-1.48	8.63	-0.01	0.0	0.00	18.167	0.03	2.96
1.4D+1.7(CL1)+0.5Lr	-2.39	18.30	-0.02	0.0	0.00	27.843	0.04	4.79
1.4D+1.7(CL2)+0.5Lr	-2.98	18.42	-0.02	0.0	0.00	27.960	0.04	5.97
1.4D+1.7(CL3)+0.5Lr	-2.35	13.58	-0.02	0.0	0.00	23.118	0.04	4.71
1.4D+1.7(CL4)+0.5Lr	-2.52	13.70	-0.02	0.0	0.00	23.239	0.04	5.05
1.4D+1.7(CL5)+0.5Lr	-2.69	18.36	-0.02	0.0	0.00	27.900	0.04	5.38
1.4D+1.7(CL6)+0.5Lr	-2.69	18.36	-0.02	0.0	0.00	27.902	0.04	5.38
1.4D+1.7(CL7)+0.5Lr	-2.44	13.64	-0.02	0.0	0.00	23.177	0.04	4.88
1.4D+1.7(CL8)+0.5Lr	-2.44	13.64	-0.02	0.0	0.00	23.177	0.05	4.88
1.2D+1.6Lr+1.0(CL1)	-3.42	17.31	-0.03	0.0	0.00	26.846	0.07	6.85
1.2D+1.6Lr+1.0(CL2)	-3.77	17.38	-0.03	0.0	0.00	26.915	0.07	7.54
1.2D+1.6Lr+1.0(CL3)	-3.40	14.53	-0.03	0.0	0.00	24.067	0.07	6.80
1.2D+1.6Lr+1.0(CL4)	-3.50	14.60	-0.03	0.0	0.00	24.138	0.07	7.00
1.2D+1.6Lr+1.0(CL5)	-3.60	17.34	-0.03	0.0	0.00	26.880	0.07	7.19
1.2D+1.6Lr+1.0(CL6)	-3.60	17.34	-0.03	0.0	0.00	26.881	0.07	7.19
1.2D+1.6Lr+1.0(CL7)	-3.45	14.56	-0.03	0.0	0.00	24.102	0.07	6.90
1.2D+1.6Lr+1.0(CL8)	-3.45	14.56	-0.03	0.0	0.00	24.102	0.07	6.90
1.2D + 1.6WL1+1.0L+0.5Lr	0.50	6.36	-0.01	0.0	0.00	15.905	0.02	1.01
1.2D + 1.6WL2+1.0L+0.5Lr	0.32	4.92	0.00	0.0	0.00	14.459	0.01	0.63
1.2D + 1.6WL3+1.0L+0.5Lr	-1.74	7.61	-0.01	0.0	0.00	17.150	0.02	3.49
1.2D + 1.6WL4+1.0L+0.5Lr	-1.92	6.51	-0.01	0.0	0.00	16.054	0.01	3.84
1.2D + 1.6WL5+1.0L+0.5Lr	-1.58	6.60	-0.01	0.0	0.00	16.140	0.02	3.16
1.2D + 1.6WL6+1.0L+0.5Lr	-1.60	4.91	-0.01	0.0	0.00	14.449	0.01	3.21
1.2D + 1.6WL7+1.0L+0.5Lr	-1.64	6.66	-0.01	0.0	0.00	16.202	0.02	3.28
1.2D + 1.6WL8+1.0L+0.5Lr	-1.73	4.92	-0.01	0.0	0.00	14.461	0.01	3.47
0.9D+1.6WL1	1.44	2.90	0.00	0.0	0.00	12.440	0.00	2.88
0.9D+1.6WL2	1.25	1.45	0.01	0.0	0.00	10.994	0.01	2.50
0.9D+1.6WL3	-0.81	4.15	0.00	0.0	0.00	13.686	0.00	1.62
0.9D+1.6WL4	-0.98	3.05	0.00	0.0	0.00	12.590	0.01	1.97
0.9D+1.6WL5	-0.65	3.14	0.00	0.0	0.00	12.675	0.00	1.29
0.9D+1.6WL6	-0.67	1.44	0.00	0.0	0.00	10.984	0.01	1.34
0.9D+1.6WL7	-0.71	3.20	0.00	0.0	0.00	12.737	0.00	1.41
0.9D+1.6WL8	-0.80	1.46	0.00	0.0	0.00	10.997	0.01	1.59

CHECK FOR BEARING PRESSURE

Load Combination & Calculation of Base pressure

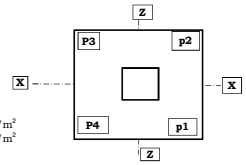
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 53.6 T/m^2 For WL/SL case = 66.1 T/m^2



Load Case	Design Load/moments			Base Pressure (T/m^2)								
	V (T)	Mz (T-m)	Mx (T-m)	V/(Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-p_{min})	Remarks
DL+Lr	18.93	4.59	0.05	6.18	5.14	0.05	13.37	13.27	3.10	2.99	11.21	O.K
DL+CL1	20.44	2.46	0.02	6.68	2.76	0.02	11.46	11.41	5.94	5.90	9.30	O.K
DL+CL2	20.51	3.16	0.02	6.70	3.53	0.02	12.25	12.21	5.19	5.14	10.09	O.K
DL+CL3	17.66	2.42	0.02	5.77	2.70	0.02	10.50	10.45	5.09	5.04	8.34	O.K
DL+CL4	17.74	2.62	0.02	5.79	2.93	0.02	10.74	10.70	4.89	4.84	8.58	O.K
DL+CL5	20.48	2.81	0.02	6.69	3.15	0.02	11.86	11.81	5.56	5.52	9.70	O.K
DL+CL6	20.48	2.81	0.02	6.69	3.15	0.02	11.86	11.81	5.56	5.52	9.70	O.K
DL+CL5	17.70	2.51	0.02	5.78	2.81	0.02	10.62	10.57	4.99	4.94	8.46	O.K
DL+CL6	17.70	2.51	0.02	5.78	2.81	0.03	10.62	10.57	4.99	4.94	8.46	O.K
DL+0.75*CL1+0.75*Lr	21.68	4.23	0.04	7.08	4.74	0.05	13.86	13.77	4.39	4.30	11.70	O.K
DL+0.75*CL2+0.75*Lr	21.73	4.75	0.04	7.10	5.32	0.05	14.46	14.37	3.82	3.73	12.30	O.K
DL+0.75*CL3+0.75*Lr	19.60	4.20	0.04	6.40	4.70	0.05	13.14	13.05	3.75	3.66	10.98	O.K
DL+0.75*CL4+0.75*Lr	19.65	4.35	0.04	6.42	4.87	0.05	13.33	13.24	3.60	3.51	11.17	O.K
DL+0.75*CL5+0.75*Lr	21.71	4.49	0.04	7.09	5.03	0.05	14.16	14.07	4.10	4.01	12.00	O.K
DL+0.75*CL6+0.75*Lr	21.71	4.49	0.04	7.09	5.03	0.05	14.16	14.07	4.10	4.01	12.00	O.K
DL+0.75*CL7+0.75*Lr	19.62	4.27	0.04	6.41	4.78	0.05	13.23	13.14	3.67	3.58	11.07	O.K
DL+0.75*CL8+0.75*Lr	19.62	4.27	0.04	6.41	4.78	0.05	13.24	13.14	3.67	3.58	11.08	O.K
DL+0.75*Lr+0.75*WL1	16.89	1.73	0.03	5.51	1.94	0.03	9.48	9.42	5.61	5.54	7.32	O.K
DL+0.75*Lr+0.75*WL2	16.21	1.91	0.03	5.29	2.13	0.03	9.46	9.40	5.19	5.13	7.30	O.K
DL+0.75*Lr+0.75*WL3	17.47	3.84	0.03	5.70	4.29	0.04	12.03	11.96	3.45	3.37	9.87	O.K
DL+0.75*Lr+0.75*WL4	16.96	4.00	0.03	5.54	4.48	0.03	12.05	11.98	3.09	3.02	9.89	O.K
DL+0.75*Lr+0.75*WL5	17.00	3.68	0.03	5.55	4.12	0.04	11.71	11.64	3.46	3.39	9.55	O.K
DL+0.75*Lr+0.75*WL6	16.20	3.71	0.03	5.29	4.15	0.03	11.47	11.41	3.17	3.11	9.31	O.K
DL+0.75*Lr+0.75*WL7	17.03	3.74	0.03	5.56	4.19	0.04	11.78	11.71	3.41	3.34	9.62	O.K
DL+0.75*Lr+0.75*WL8	16.21	3.83	0.03	5.29	4.28	0.03	11.61	11.54	3.04	2.98	9.45	O.K
0.6*DL+WL1	11.58	1.72	0.00	3.78	1.92	0.00	7.71	7.71	3.86	3.86	5.55	O.K
0.6*DL+WL2	10.68	1.49	0.01	3.49	1.66	0.01	7.16	7.14	3.83	3.82	5.00	O.K
0.6*DL+WL3	12.36	1.09	0.00	4.04	1.22	0.00	7.26	7.25	4.82	4.82	5.10	O.K
0.6*DL+WL4	11.68	1.31	0.00	3.81	1.47	0.00	7.28	7.28	4.35	4.34	5.12	O.K
0.6*DL+WL5	11.73	0.89	0.00	3.83	0.99	0.00	6.82	6.82	4.84	4.84	4.66	O.K
0.6*DL+WL6	10.67	0.91	0.00	3.49	1.02	0.00	6.51	6.51	4.46	4.46	4.35	O.K
0.6*DL+WL7	11.77	0.96	0.00	3.84	1.08	0.00	6.92	6.92	4.77	4.76	4.76	O.K
0.6*DL+WL8	10.68	1.08	0.00	3.49	1.20	0.00	6.70	6.69	4.29	4.28	4.54	O.K

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-05-2020	01-06-2020	25-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	06-05-2020	01-06-2020	25-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

CALCULATION OF DESIGN PRESSURE

Load Combination & Calculation of Base pressure

Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

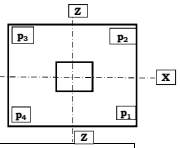
Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2

Gross SBC of soil for DL, LL= 53.6 T/m^2

For WL/SL case = 62.5 T/m^2

For WL/SL case = 66.1 T/m^2



Load Case	Design Load/moments			Base Pressure (T/m^2)							
	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-p_3)
1.4D	18.2	2.96	0.03	5.93	3.31	0.03	11.28	11.21	4.65	4.59	8.58
1.4D+1.7(CL1)+0.5Lr	27.8	4.79	0.04	9.09	5.36	0.05	16.50	16.41	5.78	5.68	13.80
1.4D+1.7(CL2)+0.5Lr	28.0	5.97	0.04	9.13	6.68	0.05	17.86	17.77	4.49	4.40	15.16
1.4D+1.7(CL3)+0.5Lr	23.1	4.71	0.04	7.55	5.27	0.05	14.87	14.77	4.32	4.23	12.17
1.4D+1.7(CL4)+0.5Lr	23.2	5.05	0.04	7.59	5.65	0.05	15.29	15.19	3.98	3.89	12.59
1.4D+1.7(CL5)+0.5Lr	27.9	5.38	0.04	9.11	6.02	0.05	17.18	17.09	5.13	5.04	14.48
1.4D+1.7(CL6)+0.5Lr	27.9	5.38	0.04	9.11	6.02	0.05	17.18	17.09	5.13	5.04	14.48
1.4D+1.7(CL7)+0.5Lr	23.2	4.88	0.04	7.57	5.46	0.05	15.08	14.98	4.15	4.06	12.38
1.4D+1.7(CL8)+0.5Lr	23.2	4.88	0.05	7.57	5.46	0.05	15.08	14.98	4.16	4.06	12.38
1.2D+1.6Lr+1.0(CL1)	26.8	6.85	0.07	8.77	7.66	0.07	18.50	18.36	3.18	3.03	15.80
1.2D+1.6Lr+1.0(CL2)	26.9	7.54	0.07	8.79	8.44	0.07	19.30	19.15	2.42	2.27	16.60
1.2D+1.6Lr+1.0(CL3)	24.1	6.80	0.07	7.86	7.61	0.07	17.54	17.40	2.32	2.17	14.84
1.2D+1.6Lr+1.0(CL4)	24.1	7.00	0.07	7.88	7.84	0.07	17.79	17.64	2.12	1.97	15.09
1.2D+1.6Lr+1.0(CL5)	26.9	7.19	0.07	8.78	8.05	0.07	18.91	18.76	2.80	2.65	16.21
1.2D+1.6Lr+1.0(CL6)	26.9	7.19	0.07	8.78	8.05	0.07	18.91	18.76	2.80	2.65	16.21
1.2D+1.6Lr+1.0(CL7)	24.1	6.90	0.07	7.87	7.72	0.07	17.67	17.52	2.22	2.07	14.97
1.2D+1.6Lr+1.0(CL8)	24.1	6.90	0.07	7.87	7.72	0.08	17.67	17.51	2.23	2.07	14.97
1.2D + 1.6WL1+1.0L+0.5Lr	15.9	1.01	0.02	5.19	1.13	0.02	8.34	8.30	6.09	6.05	5.64
1.2D + 1.6WL2+1.0L+0.5Lr	14.5	0.63	0.01	4.72	0.71	0.01	7.44	7.42	6.02	6.00	4.74
1.2D + 1.6WL3+1.0L+0.5Lr	17.2	3.49	0.02	5.60	3.90	0.02	11.53	11.48	3.72	3.67	8.83
1.2D + 1.6WL4+1.0L+0.5Lr	16.1	3.84	0.01	5.24	4.30	0.01	11.56	11.53	2.96	2.93	8.86
1.2D + 1.6WL5+1.0L+0.5Lr	16.1	3.16	0.02	5.27	3.54	0.02	10.83	10.79	3.75	3.71	8.13
1.2D + 1.6WL6+1.0L+0.5Lr	14.4	3.21	0.01	4.72	3.59	0.01	10.32	10.30	3.14	3.11	7.62
1.2D + 1.6WL7+1.0L+0.5Lr	16.2	3.28	0.02	5.29	3.68	0.02	10.99	10.94	3.64	3.59	8.29
1.2D + 1.6WL8+1.0L+0.5Lr	14.5	3.47	0.01	4.72	3.88	0.01	10.62	10.59	2.86	2.83	7.92
0.9D+1.6WL1	12.4	2.88	0.00	4.06	3.22	0.00	9.29	9.28	2.84	2.84	6.59
0.9D+1.6WL2	11.0	2.50	0.01	3.59	2.80	0.01	8.40	8.38	2.80	2.78	5.70
0.9D+1.6WL3	13.7	1.62	0.00	4.47	1.81	0.00	8.28	8.28	4.66	4.66	5.58
0.9D+1.6WL4	12.6	1.97	0.01	4.11	2.20	0.01	8.32	8.31	3.91	3.90	5.62
0.9D+1.6WL5	12.7	1.29	0.00	4.14	1.45	0.00	7.59	7.58	4.70	4.69	4.89
0.9D+1.6WL6	11.0	1.34	0.01	3.59	1.50	0.01	7.09	7.08	4.10	4.08	4.39
0.9D+1.6WL7	12.7	1.41	0.00	4.16	1.58	0.00	7.74	7.74	4.58	4.58	5.04
0.9D+1.6WL8	11.0	1.59	0.01	3.59	1.79	0.01	7.38	7.37	3.81	3.80	4.68

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-05-2020	01-06-2020	25-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

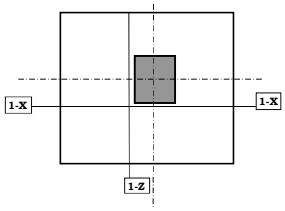
Outotec / L&T Revision No.:	0	1	2	3
Date	06-05-2020	01-06-2020	25-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.
BM1z =Moment at the face of column 1z-1Z in T-m/m.
SF(oneway) = Oneway shear force at a distance 'd' from the face of column in T/m.
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.55+2*0.5*0.201)+(0.55+2*0.5*0.201))*2
= 2.98 m
Punching Area= ((0.55+2*0.5*0.195)*(0.55+2*0.5*0.195))
= 0.56 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.3*2.5+1.2*1.8)
= 2.91 T/m^2
Area of footing, A_f= 3.06 m^2



Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 41.54 T
Strength of Concrete in shear Vc1 = 124.28 T **Table 22.6.5.2_ (a)**
Strength of Concrete in shear Vc2 = 132.28 T **Table 22.6.5.2_ (b)**
Strength of Concrete in shear Vc3 = 114.59 T **Table 22.6.5.2_ (c)**

Design Vc= 114.59 T
Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (41.54*10000)/(2980*195)
= 0.71 N/mm^2 **No shear reinforcement reqd.**

Load Case	p_{1x} (T/m^2)	$B.M_{1x}$ ™	$S.F_{(oneway)}$ (T)	$B.M_{1z}$ ™	$S.F_{(oneway)}$ (T)	$S.F_{(punching)}$ (T)	V_u/b_od	Punching check (Remarks)	Allowable shear (0.75*√(fc)/6 (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4D	8.58	1.54	3.30	1.54	3.47	21.47	0.37	OK	0.74	0.178	OK	0.41
1.4D+1.7(CL1)+0.5Lr	13.80	2.48	5.31	2.48	5.59	34.53	0.59	OK	0.74	0.287	OK	0.65
1.4D+1.7(CL2)+0.5Lr	15.16	2.73	5.84	2.73	6.14	37.94	0.65	OK	0.74	0.315	OK	0.72
1.4D+1.7(CL3)+0.5Lr	12.17	2.19	4.69	2.19	4.93	30.46	0.52	OK	0.74	0.253	OK	0.58
1.4D+1.7(CL4)+0.5Lr	12.59	2.27	4.85	2.27	5.10	31.51	0.54	OK	0.74	0.261	OK	0.60
1.4D+1.7(CL5)+0.5Lr	14.48	2.61	5.57	2.61	5.86	36.24	0.62	OK	0.74	0.301	OK	0.69
1.4D+1.7(CL6)+0.5Lr	14.48	2.61	5.57	2.61	5.86	36.24	0.62	OK	0.74	0.301	OK	0.69
1.4D+1.7(CL7)+0.5Lr	12.38	2.23	4.77	2.23	5.01	30.98	0.53	OK	0.74	0.257	OK	0.59
1.4D+1.7(CL8)+0.5Lr	12.38	2.23	4.77	2.23	5.01	30.98	0.53	OK	0.74	0.257	OK	0.59
1.2D+1.6Lr+1.0(CL1)	15.80	2.84	6.08	2.84	6.40	39.54	0.68	OK	0.74	0.328	OK	0.75
1.2D+1.6Lr+1.0(CL2)	16.60	2.99	6.39	2.99	6.72	41.54	0.71	OK	0.74	0.345	OK	0.79
1.2D+1.6Lr+1.0(CL3)	14.84	2.67	5.71	2.67	6.01	37.14	0.64	OK	0.74	0.308	OK	0.70
1.2D+1.6Lr+1.0(CL4)	15.09	2.72	5.81	2.72	6.11	37.76	0.65	OK	0.74	0.313	OK	0.71
1.2D+1.6Lr+1.0(CL5)	16.21	2.92	6.24	2.92	6.57	40.57	0.7	OK	0.74	0.337	OK	0.77
1.2D+1.6Lr+1.0(CL6)	16.21	2.92	6.24	2.92	6.57	40.57	0.7	OK	0.74	0.337	OK	0.77
1.2D+1.6Lr+1.0(CL7)	14.97	2.69	5.76	2.69	6.06	37.46	0.64	OK	0.74	0.311	OK	0.71
1.2D+1.6Lr+1.0(CL8)	14.97	2.69	5.76	2.69	6.06	37.46	0.64	OK	0.74	0.311	OK	0.71
1.2D + 1.6WL1+1.0L+0.5Lr	5.64	1.02	2.17	1.02	2.28	14.11	0.24	OK	0.74	0.117	OK	0.27
1.2D + 1.6WL2+1.0L+0.5Lr	4.74	0.85	1.82	0.85	1.92	11.86	0.2	OK	0.74	0.098	OK	0.22
1.2D + 1.6WL3+1.0L+0.5Lr	8.83	1.59	3.40	1.59	3.58	22.10	0.38	OK	0.74	0.183	OK	0.42
1.2D + 1.6WL4+1.0L+0.5Lr	8.86	1.59	3.41	1.59	3.59	22.17	0.38	OK	0.74	0.184	OK	0.42
1.2D + 1.6WL5+1.0L+0.5Lr	8.13	1.46	3.13	1.46	3.29	20.35	0.35	OK	0.74	0.169	OK	0.38
1.2D + 1.6WL6+1.0L+0.5Lr	7.62	1.37	2.93	1.37	3.09	19.07	0.33	OK	0.74	0.158	OK	0.36
1.2D + 1.6WL7+1.0L+0.5Lr	8.29	1.49	3.19	1.49	3.36	20.75	0.36	OK	0.74	0.172	OK	0.39
1.2D + 1.6WL8+1.0L+0.5Lr	7.92	1.43	3.05	1.43	3.21	19.82	0.34	OK	0.74	0.164	OK	0.37
0.9D+1.6WL1	6.59	1.19	2.54	1.19	2.67	16.49	0.28	OK	0.74	0.137	OK	0.31
0.9D+1.6WL2	5.70	1.03	2.19	1.03	2.31	14.26	0.25	OK	0.74	0.118	OK	0.27
0.9D+1.6WL3	5.58	1.00	2.15	1.00	2.26	13.96	0.24	OK	0.74	0.116	OK	0.26
0.9D+1.6WL4	5.62	1.01	2.16	1.01	2.28	14.06	0.24	OK	0.74	0.117	OK	0.27
0.9D+1.6WL5	4.89	0.88	1.88	0.88	1.98	12.24	0.21	OK	0.74	0.102	OK	0.23
0.9D+1.6WL6	4.39	0.79	1.69	0.79	1.78	10.99	0.19	OK	0.74	0.091	OK	0.21
0.9D+1.6WL7	5.04	0.91	1.94	0.91	2.04	12.61	0.22	OK	0.74	0.105	OK	0.24
0.9D+1.6WL8	4.68	0.84	1.80	0.84	1.90	11.71	0.2	OK	0.74	0.097	OK	0.22

Load Case	kn (ϕ_b=0.75)	p =	A_{st}
Ultimate Conditions			
1.4D	0.54	0.001	215
1.4D+1.7(CL1)+0.5Lr	0.87	0.002	430
1.4D+1.7(CL2)+0.5Lr	0.96	0.002	430
1.4D+1.7(CL3)+0.5Lr	0.77	0.002	430
1.4D+1.7(CL4)+0.5Lr	0.79	0.002	430
1.4D+1.7(CL5)+0.5Lr	0.91	0.002	430
1.4D+1.7(CL6)+0.5Lr	0.91	0.002	430
1.4D+1.7(CL7)+0.5Lr	0.78	0.002	430
1.4D+1.7(CL8)+0.5Lr	0.78	0.002	430
1.2D+1.6Lr+1.0(CL1)	1	0.002	430
1.2D+1.6Lr+1.0(CL2)	1.05	0.003	645
1.2D+1.6Lr+1.0(CL3)	0.94	0.002	430
1.2D+1.6Lr+1.0(CL4)	0.95	0.002	430
1.2D+1.6Lr+1.0(CL5)	1.02	0.002	430
1.2D+1.6Lr+1.0(CL6)	1.02	0.002	430
1.2D+1.6Lr+1.0(CL7)	0.94	0.002	430
1.2D+1.6Lr+1.0(CL8)	0.94	0.002	430
1.2D + 1.6WL1+1.0L+0.5Lr	0.36	0.001	215
1.2D + 1.6WL2+1.0L+0.5Lr	0.3	0.001	215
1.2D + 1.6WL3+1.0L+0.5Lr	0.56	0.001	215
1.2D + 1.6WL4+1.0L+0.5Lr	0.56	0.001	215
1.2D + 1.6WL5+1.0L+0.5Lr	0.51	0.001	215
1.2D + 1.6WL6+1.0L+0.5Lr	0.48	0.001	215
1.2D + 1.6WL7+1.0L+0.5Lr	0.52	0.001	215
1.2D + 1.6WL8+1.0L+0.5Lr	0.5	0.001	215
0.9D+1.6WL1	0.42	0.001	215
0.9D+1.6WL2	0.36	0.001	215
0.9D+1.6WL3	0.35	0.001	215
0.9D+1.6WL4	0.35	0.001	215
0.9D+1.6WL5	0.31	0.001	215
0.9D+1.6WL6	0.28	0.001	215
0.9D+1.6WL7	0.32	0.001	215
0.9D+1.6WL8	0.3	0.001	215

Location	Mux (T-m)	ϕ Kn = M_{u1}/bd^2	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing	Spacing provd.
At bottom	2.99	0.79	1.05	0.003	645.00	387.00	12φT	175.34	300	200
At top	0.00	0.00	0.00	0.0018	351.00	351.00	12φT	322.21	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**12φT**	**200**	c/c	At Top	**12φT**	**200**	c/c
Along Z Direction :	At Bottom	**12φT**	**200**	c/c	At Top	**12φT**	**200**	c/c

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-05-2020	01-06-2020	25-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

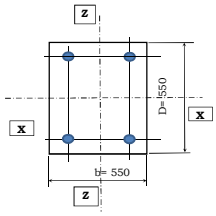
Outotec / L&T Revision No.:	0	1	2	3
Date	06-05-2020	01-06-2020	25-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

DESIGN OF COLUMN MKD C1

Input:

Clear cover to reinf.=	75mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	97.5mm
Grade of Concrete (f'_c) =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%
Pedestal Height	1700

Table 20.6.1.3.1



For Compression-controlled sections
Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupported length of column about X (l_{ux}) = 1700.00
Unsupported length of column about Z (l_{uz}) = 1700.00

For Columns not braced against sidesway,

Slenderness ratio, $K\,l_{ux} / r_x$ = 21.42 < 22 SECTION 6.2.5a
$K\,l_{uz} / r_z$ = 21.42 < 22 SECTION 6.2.5a

Where, Effective length factor K= 2
Radius of gyration r_x= 159
Radius of gyration r_z= 159

γ_y =	0.68
γ_x =	0.68

Load Combination	Factored Load/Moment (T)			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm²)
	Hx	V	Hz	Mz (Tm)	V (T)	Mx (Tm)	
1.4D	-1.48	8.63	-0.01	3.9	13.3	0.0	0.01
1.4D+1.7(CL1)+0.5Lr	-2.39	18.30	-0.02	6.3	28.2	0.1	0.03
1.4D+1.7(CL2)+0.5Lr	-2.98	18.42	-0.02	7.8	28.3	0.1	0.03
1.4D+1.7(CL3)+0.5Lr	-2.35	13.58	-0.02	6.2	20.9	0.1	0.02
1.4D+1.7(CL4)+0.5Lr	-2.52	13.70	-0.02	6.6	21.1	0.1	0.02
1.4D+1.7(CL5)+0.5Lr	-2.69	18.36	-0.02	7.0	28.2	0.1	0.03
1.4D+1.7(CL6)+0.5Lr	-2.69	18.36	-0.02	7.0	28.2	0.1	0.03
1.4D+1.7(CL7)+0.5Lr	-2.44	13.64	-0.02	6.4	21.0	0.1	0.02
1.4D+1.7(CL8)+0.5Lr	-2.44	13.64	-0.02	6.4	21.0	0.1	0.02
1.2D+1.6Lr+1.0(CL1)	-3.42	17.31	-0.03	9.0	26.6	0.1	0.03
1.2D+1.6Lr+1.0(CL2)	-3.77	17.38	-0.03	9.9	26.7	0.1	0.03
1.2D+1.6Lr+1.0(CL3)	-3.40	14.53	-0.03	8.9	22.3	0.1	0.02
1.2D+1.6Lr+1.0(CL4)	-3.50	14.60	-0.03	9.2	22.5	0.1	0.02
1.2D+1.6Lr+1.0(CL5)	-3.60	17.34	-0.03	9.4	26.7	0.1	0.03
1.2D+1.6Lr+1.0(CL6)	-3.60	17.34	-0.03	9.4	26.7	0.1	0.03
1.2D+1.6Lr+1.0(CL7)	-3.45	14.56	-0.03	9.0	22.4	0.1	0.02
1.2D+1.6Lr+1.0(CL8)	-3.45	14.56	-0.03	9.0	22.4	0.1	0.02
1.2D + 1.6WL1+1.0L+0.5Lr	0.50	6.36	-0.01	1.3	9.8	0.0	0.01
1.2D + 1.6WL2+1.0L+0.5Lr	0.32	4.92	0.00	0.8	7.6	0.0	0.01
1.2D + 1.6WL3+1.0L+0.5Lr	-1.74	7.61	-0.01	4.6	11.7	0.0	0.01
1.2D + 1.6WL4+1.0L+0.5Lr	-1.92	6.51	-0.01	5.0	10.0	0.0	0.01
1.2D + 1.6WL5+1.0L+0.5Lr	-1.58	6.60	-0.01	4.1	10.2	0.0	0.01
1.2D + 1.6WL6+1.0L+0.5Lr	-1.60	4.91	-0.01	4.2	7.6	0.0	0.01
1.2D + 1.6WL7+1.0L+0.5Lr	-1.64	6.66	-0.01	4.3	10.2	0.0	0.01
1.2D + 1.6WL8+1.0L+0.5Lr	-1.73	4.92	-0.01	4.5	7.6	0.0	0.01
0.9D+1.6WL1	1.44	2.90	0.00	3.8	4.5	0.0	0.00
0.9D+1.6WL2	1.25	1.45	0.01	3.3	2.2	0.0	0.00
0.9D+1.6WL3	-0.81	4.15	0.00	2.1	6.4	0.0	0.01
0.9D+1.6WL4	-0.98	3.05	0.00	2.6	4.7	0.0	0.00
0.9D+1.6WL5	-0.65	3.14	0.00	1.7	4.8	0.0	0.00
0.9D+1.6WL6	-0.67	1.44	0.00	1.7	2.2	0.0	0.00
0.9D+1.6WL7	-0.71	3.20	0.00	1.8	4.9	0.0	0.00
0.9D+1.6WL8	-0.80	1.46	0.00	2.1	2.2	0.0	0.00

REINFORCEMENT

Provide,

4-Nos	25 mm Dia	1964.29 Sq mm	(At Corner)	A_{sc} pvd. = 3221.43 Sq mm
2-Nos	20 mm Dia	628.57 Sq mm	(At middle)	
2-Nos	20 mm Dia	628.57 Sq mm	(At middle)	Total A_{sc} = 1.0 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85\,f'_c\,(A_g - A_{sc}) + f_y\,A_{sc}$ = 1025.65375 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.7		M_{nox}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cbD^2$						
1.4D	0.06	0.06	34.9	34.94	0.55	1.16	0.08	O.K
1.4D+1.7(CL1)+0.5Lr	0.07	0.07	40.8	40.76	0.55	1.16	0.11	O.K
1.4D+1.7(CL2)+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.18	O.K
1.4D+1.7(CL3)+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.13	O.K
1.4D+1.7(CL4)+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.15	O.K
1.4D+1.7(CL5)+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.16	O.K
1.4D+1.7(CL6)+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.16	O.K
1.4D+1.7(CL5)+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.14	O.K
1.4D+1.7(CL6)+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.14	O.K
1.2D+1.6Lr+1.0(CL1)	0.06	0.06	34.9	34.94	0.55	1.16	0.21	O.K
1.2D+1.6Lr+1.0(CL2)	0.06	0.06	34.9	34.94	0.55	1.16	0.23	O.K
1.2D+1.6Lr+1.0(CL3)	0.06	0.06	34.9	34.94	0.55	1.16	0.21	O.K
1.2D+1.6Lr+1.0(CL4)	0.06	0.06	34.9	34.94	0.55	1.16	0.21	O.K
1.2D+1.6Lr+1.0(CL5)	0.06	0.06	34.9	34.94	0.55	1.16	0.22	O.K
1.2D+1.6Lr+1.0(CL6)	0.06	0.06	34.9	34.94	0.55	1.16	0.22	O.K
1.2D+1.6Lr+1.0(CL5)	0.06	0.06	34.9	34.94	0.55	1.16	0.21	O.K
1.2D+1.6Lr+1.0(CL6)	0.06	0.06	34.9	34.94	0.55	1.16	0.21	O.K
1.2D + 1.6WL1+1.0L+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.02	O.K
1.2D + 1.6WL2+1.0L+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.01	O.K
1.2D + 1.6WL3+1.0L+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.09	O.K
1.2D + 1.6WL4+1.0L+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.11	O.K
1.2D + 1.6WL5+1.0L+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.08	O.K
1.2D + 1.6WL6+1.0L+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.09	O.K
1.2D + 1.6WL7+1.0L+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.09	O.K
1.2D + 1.6WL8+1.0L+0.5Lr	0.06	0.06	34.9	34.94	0.55	1.16	0.09	O.K
0.9D+1.6WL1	0.06	0.06	34.9	34.94	0.55	1.16	0.08	O.K
0.9D+1.6WL2	0.06	0.06	34.9	34.94	0.55	1.16	0.06	O.K
0.9D+1.6WL3	0.06	0.06	34.9	34.94	0.55	1.16	0.04	O.K
0.9D+1.6WL4	0.06	0.06	34.9	34.94	0.55	1.16	0.05	O.K
0.9D+1.6WL5	0.05	0.05	29.1	29.12	0.55	1.16	0.04	O.K
0.9D+1.6WL6	0.05	0.05	29.1	29.12	0.55	1.16	0.04	O.K
0.9D+1.6WL7	0.05	0.05	29.1	29.12	0.55	1.16	0.04	O.K
0.9D+1.6WL8	0.05	0.05	29.1	29.12	0.55	1.16	0.05	O.K

Links, **10mm dia @ 300 C/C**

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	06-05-2020	01-06-2020	25-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	06-05-2020	01-06-2020	25-10-2023	
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3.3 Design of Foundation mkd. F3 (GRID D2 & A2)

	GRID D2 (A2 SIMILAR)		
Load Combinations	FX (T)	FY (T)	FZ (T)
1 DL	0.8	5.9	-0.2
2 LLROOF	0.9	2.7	-0.1
3 CL1(L+XS)	0.2	2.1	-0.1
4 CL2(L-XS)	0.1	1.9	-0.1
5 CL3(R+XS)	0.5	4.8	-0.2
6 CL4(R-XS)	0.2	4.7	-0.2
7 CL5(L+ZT)	0.2	1.8	0.0
8 CL6(L-ZT)	0.2	2.2	-0.2
9 CL7(R+ZT)	0.3	4.6	-0.1
10 CL8(R-ZT)	0.3	4.9	-0.3
21 WL1(+X)S	0.1	-0.6	0.0
22 WL2(+X)P	0.2	-1.4	0.1
23 WL3(-X)S	-1.3	-1.4	0.1
24 WL4(-X)P	-1.1	-2.2	0.1
25 WL5(+Z)S	0.0	-4.7	1.3
26 WL6(+Z)P	0.1	-5.2	1.2
27 WL7(-Z)S	-0.1	2.1	-1.2
28 WL8(-Z)P	0.1	0.8	-1.0

Strength Design Load Combinations	GRID D2		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4D	1.17	8.32	-0.35
1.4D+1.7(CL1)+0.5Lr	2.04	13.17	-0.58
1.4D+1.7(CL2)+0.5Lr	1.86	12.95	-0.55
1.4D+1.7(CL3)+0.5Lr	2.49	17.80	-0.77
1.4D+1.7(CL4)+0.5Lr	1.92	17.65	-0.75
1.4D+1.7(CL5)+0.5Lr	1.95	12.77	-0.43
1.4D+1.7(CL6)+0.5Lr	1.95	13.35	-0.69
1.4D+1.7(CL5)+0.5Lr	2.20	17.44	-0.63
1.4D+1.7(CL6)+0.5Lr	2.20	18.01	-0.90
1.2D+1.6Lr+1.0(CL1)	2.68	13.48	-0.57
1.2D+1.6Lr+1.0(CL2)	2.58	13.35	-0.55
1.2D+1.6Lr+1.0(CL3)	2.95	16.20	-0.69
1.2D+1.6Lr+1.0(CL4)	2.61	16.11	-0.67
1.2D+1.6Lr+1.0(CL5)	2.63	13.24	-0.48
1.2D+1.6Lr+1.0(CL6)	2.63	13.58	-0.64
1.2D+1.6Lr+1.0(CL7)	2.78	15.99	-0.60
1.2D+1.6Lr+1.0(CL8)	2.78	16.33	-0.76
1.2D + 1.6WL1+1.0L+0.5Lr	1.57	7.57	-0.33
1.2D + 1.6WL2+1.0L+0.5Lr	1.83	6.24	-0.26
1.2D + 1.6WL3+1.0L+0.5Lr	-0.61	6.16	-0.22
1.2D + 1.6WL4+1.0L+0.5Lr	-0.35	4.93	-0.16
1.2D + 1.6WL5+1.0L+0.5Lr	1.37	1.01	1.72
1.2D + 1.6WL6+1.0L+0.5Lr	1.56	0.17	1.57
1.2D + 1.6WL7+1.0L+0.5Lr	1.31	11.76	-2.27
1.2D + 1.6WL8+1.0L+0.5Lr	1.65	9.77	-1.98
0.9D+1.6WL1	0.88	4.45	-0.20
0.9D+1.6WL2	1.14	3.12	-0.13
0.9D+1.6WL3	-1.31	3.04	-0.10
0.9D+1.6WL4	-1.05	1.81	-0.03
0.9D+1.6WL5	0.68	-2.11	1.85
0.9D+1.6WL6	0.86	-2.95	1.70
0.9D+1.6WL7	0.61	8.64	-2.14
0.9D+1.6WL8	0.95	6.65	-1.85

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Ma'aden Revision No:	A01	C01	D01
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Outotec / L&T Revision No.:	0	1	2	3
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DESIGN OF FOUNDATION MKD. F3

	Along-X	Along-Z
Size of column :	0.55m	0.55m
Size of fdn. raft :	1.75m	1.75m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W
Crane Load	CL

Design Input:

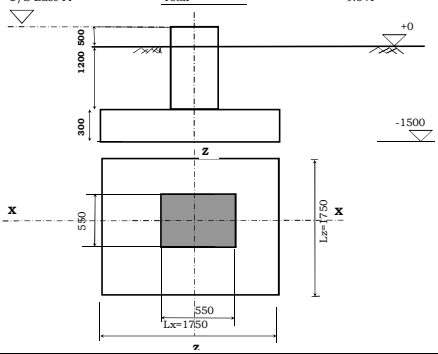
Characteristic strength of concrete, f 'c=	35 MPa
Yield stress of steel reinforcement, fy=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	20mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	20mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Calculation of Weight of Foundation :

Weight of raft & Col.= (1.75*1.75*0.3*2.5)+(0.55*0.55*1.7*2.5) = 3.58T

Weight of soil = ((1.75*1.75)-(0.55*0.55))*1.2*1.8 = 5.96T

U/S Base Pl Total = 9.54T

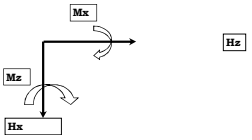


Effective depth of slab along X direction = 215mm = 0.215m

Effective depth of slab along Z direction = 195mm = 0.195m

Clear projection along Z direction = 0.6m

Clear projection along X direction = 0.6m



WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+Lr	1.73	8.62	-0.35	0.0	0.00	18.16	1.73	0.70	-0.35	3.46
DL+CL1	1.08	8.01	-0.35	0.0	0.00	17.55	1.08	0.70	-0.35	2.17
DL+CL2	0.98	7.88	-0.34	0.0	0.00	17.42	0.98	0.67	-0.34	1.96
DL+CL3	1.35	10.73	-0.47	0.0	0.00	20.27	1.35	0.94	-0.47	2.70
DL+CL4	1.02	10.64	-0.46	0.0	0.00	20.18	1.02	0.91	-0.46	2.03
DL+CL5	1.03	7.77	-0.27	0.0	0.00	17.31	1.03	0.53	-0.27	2.06
DL+CL6	1.03	8.11	-0.42	0.0	0.00	17.65	1.03	0.84	-0.42	2.06
DL+CL7	1.18	10.52	-0.38	0.0	0.00	20.06	1.18	0.77	-0.38	2.36
DL+CL8	1.18	10.86	-0.54	0.0	0.00	20.40	1.18	1.08	-0.54	2.36
DL+0.75*CL1+0.75*Lr	1.69	9.50	-0.40	0.0	0.00	19.04	1.69	0.81	-0.40	3.39
DL+0.75*CL2+0.75*Lr	1.61	9.40	-0.39	0.0	0.00	18.94	1.61	0.78	-0.39	3.23
DL+0.75*CL3+0.75*Lr	1.89	11.54	-0.49	0.0	0.00	21.08	1.89	0.98	-0.49	3.78
DL+0.75*CL4+0.75*Lr	1.64	11.47	-0.48	0.0	0.00	21.01	1.64	0.96	-0.48	3.28
DL+0.75*CL5+0.75*Lr	1.65	9.32	-0.34	0.0	0.00	18.86	1.65	0.68	-0.34	3.31
DL+0.75*CL6+0.75*Lr	1.65	9.58	-0.46	0.0	0.00	19.12	1.65	0.91	-0.46	3.31
DL+0.75*CL7+0.75*Lr	1.77	11.38	-0.43	0.0	0.00	20.92	1.77	0.86	-0.43	3.53
DL+0.75*CL8+0.75*Lr	1.77	11.63	-0.55	0.0	0.00	21.17	1.77	1.09	-0.55	3.53
DL+0.75*Lr+0.75*WL1	1.57	7.53	-0.32	0.0	0.00	17.07	1.57	0.63	-0.32	3.13
DL+0.75*Lr+0.75*WL2	1.69	6.90	-0.29	0.0	0.00	16.44	1.69	0.57	-0.29	3.37
DL+0.75*Lr+0.75*WL3	0.54	6.87	-0.27	0.0	0.00	16.41	0.54	0.53	-0.27	1.08
DL+0.75*Lr+0.75*WL4	0.66	6.29	-0.24	0.0	0.00	15.83	0.66	0.47	-0.24	1.33
DL+0.75*Lr+0.75*WL5	1.47	4.45	0.64	0.0	0.00	13.99	1.47	1.29	0.64	2.94
DL+0.75*Lr+0.75*WL6	1.56	4.06	0.58	0.0	0.00	13.60	1.56	1.15	0.58	3.12
DL+0.75*Lr+0.75*WL7	1.44	9.49	-1.23	0.0	0.00	19.03	1.44	2.45	-1.23	2.88
DL+0.75*Lr+0.75*WL8	1.60	8.56	-1.09	0.0	0.00	18.10	1.60	2.18	-1.09	3.20
0.6*DL+WL1	0.58	3.00	-0.14	0.0	0.00	12.54	0.58	0.27	-0.14	1.16
0.6*DL+WL2	0.74	2.18	-0.09	0.0	0.00	11.72	0.74	0.19	-0.09	1.48
0.6*DL+WL3	-0.79	2.12	-0.07	0.0	0.00	11.66	-0.79	0.14	-0.07	1.58
0.6*DL+WL4	-0.62	1.35	-0.03	0.0	0.00	10.89	-0.62	0.06	-0.03	1.25
0.6*DL+WL5	0.45	-1.10	1.14	0.0	0.00	8.44	0.45	2.29	1.14	0.91
0.6*DL+WL6	0.57	-1.62	1.05	0.0	0.00	7.92	0.57	2.11	1.05	1.14
0.6*DL+WL7	0.41	5.62	-1.35	0.0	0.00	15.16	0.41	2.70	-1.35	0.83
0.6*DL+WL8	0.63	4.38	-1.17	0.0	0.00	13.92	0.63	2.33	-1.17	1.25

Check for overturning and sliding:

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
DL+Lr	18.16	1.73	3.46	-0.35	0.70	4.59	22.55	4.19	20.61
DL+CL1	17.55	1.08	2.17	-0.35	0.70	7.08	21.79	6.47	19.92
DL+CL2	17.42	0.98	1.96	-0.34	0.67	7.79	22.72	7.13	20.77
DL+CL3	20.27	1.35	2.70	-0.47	0.94	6.58	18.95	6.01	17.33
DL+CL4	20.18	1.02	2.03	-0.46	0.91	8.70	19.41	7.95	17.74
DL+CL5	17.31	1.03	2.06	-0.27	0.53	7.35	28.56	6.72	26.11
DL+CL6	17.65	1.03	2.06	-0.42	0.84	7.49	18.34	6.85	16.76
DL+CL7	20.06	1.18	2.36	-0.38	0.77	7.43	22.88	6.79	20.92
DL+CL8	20.40	1.18	2.36	-0.54	1.08	7.55	16.54	6.90	15.12
DL+0.75*CL1+0.75*Lr	19.04	1.69	3.39	-0.40	0.81	4.92	20.57	4.50	18.80
DL+0.75*CL2+0.75*Lr	18.94	1.61	3.23	-0.39	0.78	5.14	21.12	4.70	19.31
DL+0.75*CL3+0.75*Lr	21.08	1.89	3.78	-0.49	0.98	4.88	18.76	4.46	17.15
DL+0.75*CL4+0.75*Lr	21.01	1.64	3.28	-0.48	0.96	5.60	19.08	5.12	17.44
DL+0.75*CL5+0.75*Lr	18.86	1.65	3.31	-0.34	0.68	4.99	24.30	4.56	22.21
DL+0.75*CL6+0.75*Lr	19.12	1.65	3.31	-0.46	0.91	5.06	18.31	4.62	16.75
DL+0.75*CL7+0.75*Lr	20.92	1.77	3.53	-0.43	0.86	5.18	21.37	4.74	19.54
DL+0.75*CL8+0.75*Lr	21.17	1.77	3.53	-0.55	1.09	5.24	16.99	4.80	15.53
DL+0.75*Lr+0.75*WL1	17.07	1.57	3.13	-0.32	0.63	4.77	23.54	4.36	21.52
DL+0.75*Lr+0.75*WL2	16.44	1.69	3.37	-0.29	0.57	4.26	25.24	3.90	23.08
DL+0.75*Lr+0.75*WL3	16.41	0.54	1.08	-0.27	0.53	13.30	26.84	12.16	24.54
DL+0.75*Lr+0.75*WL4	15.83	0.66	1.33	-0.24	0.47	10.43	29.31	9.54	26.80
DL+0.75*Lr+0.75*WL5	13.99	1.47	2.94	0.64	1.29	4.16	9.52	3.80	8.70
DL+0.75*Lr+0.75*WL6	13.60	1.56	3.12	0.58	1.15	3.81	10.33	3.49	9.45
DL+0.75*Lr+0.75*WL7	19.03	1.44	2.88	-1.23	2.45	5.78	6.79	5.28	6.21
DL+0.75*Lr+0.75*WL8	18.10	1.60	3.20	-1.09	2.18	4.94	7.27	4.52	6.64
0.6*DL+WL1	12.54	0.58	1.16	-0.14	0.27	9.48	40.20	8.67	36.76
0.6*DL+WL2	11.72	0.74	1.48	-0.09	0.19	6.91	54.76	6.32	50.07
0.6*DL+WL3	11.66	-0.79	1.58	-0.07	0.14	6.47	72.70	5.92	66.47
0.6*DL+WL4	10.89	-0.62	1.25	-0.03	0.06	7.64	166.63	6.99	152.34
0.6*DL+WL5	8.44	0.45	0.91	1.14	2.29	8.13	3.23	7.44	2.95
0.6*DL+WL6	7.92	0.57	1.14	1.05	2.11	6.07	3.29	5.55	3.01
0.6*DL+WL7	15.16	0.41	0.83	-1.35	2.70	16.03	4.92	14.65	4.50
0.6*DL+WL8	13.92	0.63	1.25	-1.17	2.33	9.71	5.22	8.88	4.77

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-05-2020	01-06-2020	25-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	06-05-2020	01-06-2020	25-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

ULTIMATE STRESS DESIGN	Load at U/S Of Base Pit					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4D	1.17	8.32	-0.35	0.0	0.00	17.856	0.69	2.34
1.4D+1.7(CL1)+0.5Lr	2.04	13.17	-0.58	0.0	0.00	22.707	1.15	4.08
1.4D+1.7(CL2)+0.5Lr	1.86	12.95	-0.55	0.0	0.00	22.488	1.10	3.72
1.4D+1.7(CL3)+0.5Lr	2.49	17.80	-0.77	0.0	0.00	27.339	1.55	4.98
1.4D+1.7(CL4)+0.5Lr	1.92	17.65	-0.75	0.0	0.00	27.191	1.50	3.84
1.4D+1.7(CL5)+0.5Lr	1.95	12.77	-0.43	0.0	0.00	22.309	0.86	3.90
1.4D+1.7(CL6)+0.5Lr	1.95	13.35	-0.69	0.0	0.00	22.886	1.39	3.90
1.4D+1.7(CL5)+0.5Lr	2.20	17.44	-0.63	0.0	0.00	26.977	1.26	4.41
1.4D+1.7(CL6)+0.5Lr	2.20	18.01	-0.90	0.0	0.00	27.554	1.79	4.41
1.2D+1.6Lr+1.0(CL1)	2.68	13.48	-0.57	0.0	0.00	23.015	1.14	5.37
1.2D+1.6Lr+1.0(CL2)	2.58	13.35	-0.55	0.0	0.00	22.887	1.11	5.15
1.2D+1.6Lr+1.0(CL3)	2.95	16.20	-0.69	0.0	0.00	25.740	1.37	5.89
1.2D+1.6Lr+1.0(CL4)	2.61	16.11	-0.67	0.0	0.00	25.653	1.35	5.23
1.2D+1.6Lr+1.0(CL5)	2.63	13.24	-0.48	0.0	0.00	22.781	0.97	5.26
1.2D+1.6Lr+1.0(CL6)	2.63	13.58	-0.64	0.0	0.00	23.121	1.28	5.26
1.2D+1.6Lr+1.0(CL7)	2.78	15.99	-0.60	0.0	0.00	25.527	1.20	5.56
1.2D+1.6Lr+1.0(CL8)	2.78	16.33	-0.76	0.0	0.00	25.866	1.51	5.56
1.2D + 1.6WL1+1.0L+0.5Lr	1.57	7.57	-0.33	0.0	0.00	17.109	0.66	3.15
1.2D + 1.6WL2+1.0L+0.5Lr	1.83	6.24	-0.26	0.0	0.00	15.784	0.52	3.67
1.2D + 1.6WL3+1.0L+0.5Lr	-0.61	6.16	-0.22	0.0	0.00	15.703	0.45	1.23
1.2D + 1.6WL4+1.0L+0.5Lr	-0.35	4.93	-0.16	0.0	0.00	14.467	0.32	0.70
1.2D + 1.6WL5+1.0L+0.5Lr	1.37	1.01	1.72	0.0	0.00	10.551	3.44	2.75
1.2D + 1.6WL6+1.0L+0.5Lr	1.56	0.17	1.57	0.0	0.00	9.712	3.15	3.12
1.2D + 1.6WL7+1.0L+0.5Lr	1.31	11.76	-2.27	0.0	0.00	21.302	4.54	2.62
1.2D + 1.6WL8+1.0L+0.5Lr	1.65	9.77	-1.98	0.0	0.00	19.307	3.96	3.30
0.9D+1.6WL1	0.88	4.45	-0.20	0.0	0.00	13.989	0.41	1.75
0.9D+1.6WL2	1.14	3.12	-0.13	0.0	0.00	12.664	0.27	2.27
0.9D+1.6WL3	-1.31	3.04	-0.10	0.0	0.00	12.583	0.20	2.62
0.9D+1.6WL4	-1.05	1.81	-0.03	0.0	0.00	11.348	0.06	2.10
0.9D+1.6WL5	0.68	-2.11	1.85	0.0	0.00	7.431	3.69	1.35
0.9D+1.6WL6	0.86	-2.95	1.70	0.0	0.00	6.593	3.40	1.73
0.9D+1.6WL7	0.61	8.64	-2.14	0.0	0.00	18.183	4.28	1.22
0.9D+1.6WL8	0.95	6.65	-1.85	0.0	0.00	16.188	3.70	1.91

Check for Bearing Pressure

Load Combination & Calculation of Base pressure

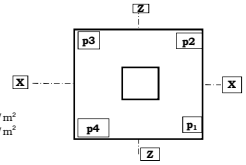
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 53.6 T/m^2 For WL/SL case = 66.1 T/m^2



Load Case	Design Load/moments			Base Pressure (T/m^2)								
	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2.Lz$	6.Mx/ $Lx.(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-p_1)	Remarks
DL+Lr	18.16	3.46	0.70	5.93	3.88	0.79	12.59	11.02	4.84	3.26	10.43	O.K
DL+CL1	17.55	2.17	0.70	5.73	2.43	0.79	10.95	9.37	6.09	4.51	8.79	O.K
DL+CL2	17.42	1.96	0.67	5.69	2.19	0.75	10.63	9.13	6.25	4.75	8.47	O.K
DL+CL3	20.27	2.70	0.94	6.62	3.02	1.05	12.69	10.59	6.65	4.55	10.53	O.K
DL+CL4	20.18	2.03	0.91	6.59	2.27	1.02	11.88	9.85	7.34	5.30	9.72	O.K
DL+CL5	17.31	2.06	0.53	5.65	2.31	0.59	10.56	9.37	5.94	4.75	8.40	O.K
DL+CL6	17.65	2.06	0.84	5.76	2.31	0.94	11.02	9.13	6.40	4.51	8.86	O.K
DL+CL7	20.06	2.36	0.77	6.55	2.65	0.86	12.05	10.34	6.76	5.05	9.89	O.K
DL+CL8	20.40	2.36	1.08	6.66	2.65	1.21	12.51	10.10	7.22	4.81	10.35	O.K
DL+0.75*CL1+0.75*Lr	19.04	3.39	0.81	6.22	3.79	0.91	12.91	11.10	5.33	3.52	10.75	O.K
DL+0.75*CL2+0.75*Lr	18.94	3.23	0.78	6.18	3.61	0.88	12.68	10.92	5.45	3.69	10.52	O.K
DL+0.75*CL3+0.75*Lr	21.08	3.78	0.98	6.88	4.23	1.10	14.22	12.02	5.75	3.55	12.06	O.K
DL+0.75*CL4+0.75*Lr	21.01	3.28	0.96	6.86	3.68	1.08	13.62	11.46	6.27	4.11	11.46	O.K
DL+0.75*CL5+0.75*Lr	18.86	3.31	0.68	6.16	3.70	0.76	12.62	11.10	5.22	3.70	10.46	O.K
DL+0.75*CL6+0.75*Lr	19.12	3.31	0.91	6.24	3.70	1.02	12.97	10.92	5.56	3.52	10.81	O.K
DL+0.75*CL7+0.75*Lr	20.92	3.53	0.86	6.83	3.96	0.96	13.75	11.83	5.84	3.92	11.59	O.K
DL+0.75*CL8+0.75*Lr	21.17	3.53	1.09	6.91	3.96	1.22	14.09	11.65	6.18	3.74	11.93	O.K
DL+0.75*Lr+0.75*WL1	17.07	3.13	0.63	5.57	3.51	0.71	11.79	10.37	4.78	3.36	9.63	O.K
DL+0.75*Lr+0.75*WL2	16.44	3.37	0.57	5.37	3.78	0.64	11.79	10.51	4.23	2.95	9.63	O.K
DL+0.75*Lr+0.75*WL3	16.41	1.08	0.53	5.36	1.21	0.60	9.16	7.97	6.75	5.55	7.00	O.K
DL+0.75*Lr+0.75*WL4	15.83	1.33	0.47	5.17	1.49	0.53	9.18	8.13	6.21	5.15	7.02	O.K
DL+0.75*Lr+0.75*WL5	13.99	2.94	1.29	4.57	3.30	1.44	11.30	8.42	4.71	1.83	9.14	O.K
DL+0.75*Lr+0.75*WL6	13.60	3.12	1.15	4.44	3.49	1.29	11.22	8.64	4.24	1.66	9.06	O.K
DL+0.75*Lr+0.75*WL7	19.03	2.88	2.45	6.21	3.23	2.74	14.19	8.70	7.73	2.24	12.03	O.K
DL+0.75*Lr+0.75*WL8	18.10	3.20	2.18	5.91	3.59	2.44	13.93	9.06	6.76	1.88	11.77	O.K
0.6*DL+WL1	12.54	1.16	0.27	4.10	1.30	0.31	7.70	7.09	5.11	4.49	5.54	O.K
0.6*DL+WL2	11.72	1.48	0.19	3.83	1.66	0.21	7.70	7.28	4.37	3.96	5.54	O.K
0.6*DL+WL3	11.66	1.58	0.14	3.81	1.77	0.16	7.73	7.42	4.20	3.89	5.57	O.K
0.6*DL+WL4	10.89	1.25	0.06	3.56	1.40	0.06	7.02	6.89	4.22	4.10	4.86	O.K
0.6*DL+WL5	8.44	0.91	2.29	2.76	1.02	2.56	8.34	3.21	6.30	1.18	6.18	O.K
0.6*DL+WL6	7.92	1.14	2.11	2.59	1.28	2.36	8.23	3.50	5.67	0.95	6.07	O.K
0.6*DL+WL7	15.16	0.83	2.70	4.95	0.93	3.02	10.90	4.86	9.04	3.01	8.74	O.K
0.6*DL+WL8	13.92	1.25	2.33	4.54	1.40	2.61	10.56	5.34	7.75	2.53	8.40	O.K

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-05-2020	01-06-2020	25-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	06-05-2020	01-06-2020	25-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

Calculation of Design Pressure

Load Combination & Calculation of Base pressure

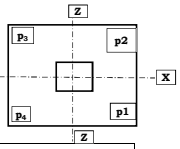
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 — For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 53.6 T/m^2 — For WL/SL case = 66.1 T/m^2



Load Case	Design Load/moments			Base Pressure (T/m^2)							
	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-p_{min})
1.4D	17.9	2.34	0.69	5.83	2.62	0.77	11.23	9.68	5.98	4.43	8.53
1.4D+1.7(CL1)+0.5Lr	22.7	4.08	1.15	7.41	4.57	1.29	15.27	12.69	6.14	3.56	12.57
1.4D+1.7(CL2)+0.5Lr	22.5	3.72	1.10	7.34	4.16	1.23	14.73	12.27	6.41	3.95	12.03
1.4D+1.7(CL3)+0.5Lr	27.3	4.98	1.55	8.93	5.57	1.73	18.23	14.76	7.09	3.62	15.53
1.4D+1.7(CL4)+0.5Lr	27.2	3.84	1.50	8.88	4.30	1.68	16.87	13.50	8.26	4.89	14.17
1.4D+1.7(CL5)+0.5Lr	22.3	3.90	0.86	7.28	4.36	0.96	14.61	12.69	5.88	3.96	11.91
1.4D+1.7(CL6)+0.5Lr	22.9	3.90	1.39	7.47	4.36	1.56	15.39	12.28	6.67	3.55	12.69
1.4D+1.7(CL5)+0.5Lr	27.0	4.41	1.26	8.81	4.94	1.41	17.16	14.33	7.28	4.46	14.46
1.4D+1.7(CL6)+0.5Lr	27.6	4.41	1.79	9.00	4.94	2.01	17.94	13.93	8.07	4.05	15.24
1.2D+1.6Lr+1.0(CL1)	23.0	5.37	1.14	7.52	6.01	1.28	16.80	14.25	4.79	2.23	14.10
1.2D+1.6Lr+1.0(CL2)	22.9	5.15	1.11	7.47	5.77	1.24	16.48	14.00	4.94	2.47	13.78
1.2D+1.6Lr+1.0(CL3)	25.7	5.89	1.37	8.40	6.60	1.54	18.54	15.47	5.34	2.27	15.84
1.2D+1.6Lr+1.0(CL4)	25.7	5.23	1.35	8.38	5.85	1.51	17.74	14.72	6.03	3.02	15.04
1.2D+1.6Lr+1.0(CL5)	22.8	5.26	0.97	7.44	5.89	1.08	16.41	14.24	4.63	2.47	13.71
1.2D+1.6Lr+1.0(CL6)	23.1	5.26	1.28	7.55	5.89	1.43	16.87	14.01	5.09	2.23	14.17
1.2D+1.6Lr+1.0(CL7)	25.5	5.56	1.20	8.34	6.23	1.35	17.91	15.21	5.46	2.76	15.21
1.2D+1.6Lr+1.0(CL8)	25.9	5.56	1.51	8.45	6.23	1.70	18.37	14.98	5.92	2.53	15.67
1.2D + 1.6WL1+1.0L+0.5Lr	17.1	3.15	0.66	5.59	3.53	0.74	11.85	10.37	4.80	3.32	9.15
1.2D + 1.6WL2+1.0L+0.5Lr	15.8	3.67	0.52	5.15	4.11	0.59	11.85	10.68	3.63	2.46	9.15
1.2D + 1.6WL3+1.0L+0.5Lr	15.7	1.23	0.45	5.13	1.37	0.50	9.00	8.00	6.26	5.25	6.30
1.2D + 1.6WL4+1.0L+0.5Lr	14.5	0.70	0.32	4.72	0.78	0.35	7.86	7.15	6.29	5.59	5.16
1.2D + 1.6WL5+1.0L+0.5Lr	10.6	2.75	3.44	3.45	3.08	3.85	12.37	4.68	6.21	-1.48	9.67
1.2D + 1.6WL6+1.0L+0.5Lr	9.7	3.12	3.15	3.17	3.50	3.53	12.20	5.14	5.20	-1.85	9.50
1.2D + 1.6WL7+1.0L+0.5Lr	21.3	2.62	4.54	6.96	2.93	5.08	16.97	6.81	11.10	0.94	14.27
1.2D + 1.6WL8+1.0L+0.5Lr	19.3	3.30	3.96	6.30	3.70	4.43	16.43	7.57	9.03	0.18	13.73
0.9D+1.6WL1	14.0	1.75	0.41	4.57	1.96	0.46	8.99	8.07	5.06	4.15	6.29
0.9D+1.6WL2	12.7	2.27	0.27	4.14	2.54	0.30	8.98	8.38	3.89	3.29	6.28
0.9D+1.6WL3	12.6	2.62	0.20	4.11	2.94	0.22	9.26	8.83	3.39	2.95	6.56
0.9D+1.6WL4	11.3	2.10	0.06	3.71	2.35	0.07	8.12	7.98	3.43	3.29	5.42
0.9D+1.6WL5	7.4	1.35	3.69	2.43	1.51	4.13	10.07	1.81	7.04	-1.22	7.37
0.9D+1.6WL6	6.6	1.73	3.40	2.15	1.93	3.81	9.90	2.28	6.03	-1.59	7.20
0.9D+1.6WL7	18.2	1.22	4.28	5.94	1.37	4.80	14.10	4.51	11.36	1.77	11.40
0.9D+1.6WL8	16.2	1.91	3.70	5.29	2.13	4.14	13.56	5.28	9.30	1.01	10.86

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-05-2020	01-06-2020	25-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	06-05-2020	01-06-2020	25-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
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Approved	PRI	PRI	PRI	

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

$SF_{(oneway)}$ = Oneway shear force at a distance 'd' from the face of column in T/m.

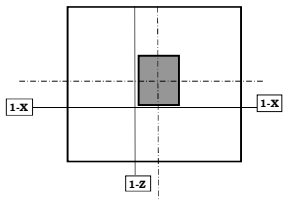
$SF_{(punching)}$ = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.55+2*0.5*0.201)+(0.55+2*0.5*0.201))*2

= 2.98 m

Punching Area= ((0.55+2*0.5*0.195)*(0.55+2*0.5*0.195))

= 0.56 m^2 Punching shear stress to be checked



Downward pressure due to self weight of raft and soil = (0.3*2.5+1.2*1.8)

= 2.91 T/m^2

Area of footing, A_f= 3.06 m^2

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 39.64 T

Strength of Concrete in shear Vc1 = 124.28 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 132.28 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 114.59 T **Table 22.6.5.2_ (c)**

Design Vc= 114.59 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (39.64*10000)/(2980*195)

= 0.68 N/mm^2 **No shear reinforcement reqd.**

Load Case	p_{1x} (T)	B.M_{1x} (T-m)	S.$F_{(Oneway)}$ (T)	B.M_{1z} ™	S.$F_{(oneway)}$ (T)	S.$F_{(punching)}$ (T)	V_u/b_0d	Punching check (Remarks)	Allowable shear (0.75*√(fc)/6 (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4D	8.53	1.54	3.28	1.54	3.45	21.35	0.37	OK	0.74	0.177	OK	0.40
1.4D+1.7(CL1)+0.5Lr	12.57	2.26	4.84	2.26	5.09	31.46	0.54	OK	0.74	0.261	OK	0.60
1.4D+1.7(CL2)+0.5Lr	12.03	2.17	4.63	2.17	4.87	30.11	0.52	OK	0.74	0.25	OK	0.57
1.4D+1.7(CL3)+0.5Lr	15.53	2.80	5.98	2.80	6.29	38.86	0.67	OK	0.74	0.323	OK	0.74
1.4D+1.7(CL4)+0.5Lr	14.17	2.55	5.46	2.55	5.74	35.46	0.61	OK	0.74	0.294	OK	0.67
1.4D+1.7(CL5)+0.5Lr	11.91	2.14	4.59	2.14	4.82	29.80	0.51	OK	0.74	0.247	OK	0.56
1.4D+1.7(CL6)+0.5Lr	12.69	2.28	4.89	2.28	5.14	31.76	0.55	OK	0.74	0.264	OK	0.60
1.4D+1.7(CL5)+0.5Lr	14.46	2.60	5.57	2.60	5.86	36.19	0.62	OK	0.74	0.3	OK	0.68
1.4D+1.7(CL6)+0.5Lr	15.24	2.74	5.87	2.74	6.17	38.14	0.66	OK	0.74	0.317	OK	0.72
1.2D+1.6Lr+1.0(CL1)	14.10	2.54	5.43	2.54	5.71	35.29	0.61	OK	0.74	0.293	OK	0.67
1.2D+1.6Lr+1.0(CL2)	13.78	2.48	5.31	2.48	5.58	34.48	0.59	OK	0.74	0.286	OK	0.65
1.2D+1.6Lr+1.0(CL3)	15.84	2.85	6.10	2.85	6.42	39.64	0.68	OK	0.74	0.329	OK	0.75
1.2D+1.6Lr+1.0(CL4)	15.04	2.71	5.79	2.71	6.09	37.64	0.65	OK	0.74	0.312	OK	0.71
1.2D+1.6Lr+1.0(CL5)	13.71	2.47	5.28	2.47	5.55	34.31	0.59	OK	0.74	0.285	OK	0.65
1.2D+1.6Lr+1.0(CL6)	14.17	2.55	5.46	2.55	5.74	35.46	0.61	OK	0.74	0.294	OK	0.67
1.2D+1.6Lr+1.0(CL7)	15.21	2.74	5.86	2.74	6.16	38.06	0.66	OK	0.74	0.316	OK	0.72
1.2D+1.6Lr+1.0(CL8)	15.67	2.82	6.03	2.82	6.35	39.21	0.67	OK	0.74	0.325	OK	0.74
1.2D + 1.6WL1+1.0L+0.5Lr	9.15	1.65	3.52	1.65	3.71	22.90	0.39	OK	0.74	0.19	OK	0.43
1.2D + 1.6WL2+1.0L+0.5Lr	9.15	1.65	3.52	1.65	3.71	22.90	0.39	OK	0.74	0.19	OK	0.43
1.2D + 1.6WL3+1.0L+0.5Lr	6.30	1.13	2.43	1.13	2.55	15.77	0.27	OK	0.74	0.131	OK	0.30
1.2D + 1.6WL4+1.0L+0.5Lr	5.16	0.93	1.99	0.93	2.09	12.91	0.22	OK	0.74	0.107	OK	0.24
1.2D + 1.6WL5+1.0L+0.5Lr	9.67	1.74	3.72	1.74	3.92	24.20	0.42	OK	0.74	0.201	OK	0.46
1.2D + 1.6WL6+1.0L+0.5Lr	9.50	1.71	3.66	1.71	3.85	23.77	0.41	OK	0.74	0.197	OK	0.45
1.2D + 1.6WL7+1.0L+0.5Lr	14.27	2.57	5.49	2.57	5.78	35.71	0.61	OK	0.74	0.296	OK	0.68
1.2D + 1.6WL8+1.0L+0.5Lr	13.73	2.47	5.29	2.47	5.56	34.36	0.59	OK	0.74	0.285	OK	0.65
0.9D+1.6WL1	6.29	1.13	2.42	1.13	2.55	15.74	0.27	OK	0.74	0.131	OK	0.30
0.9D+1.6WL2	6.28	1.13	2.42	1.13	2.54	15.72	0.27	OK	0.74	0.13	OK	0.30
0.9D+1.6WL3	6.56	1.18	2.53	1.18	2.66	16.42	0.28	OK	0.74	0.136	OK	0.31
0.9D+1.6WL4	5.42	0.98	2.09	0.98	2.20	13.56	0.23	OK	0.74	0.113	OK	0.26
0.9D+1.6WL5	7.37	1.33	2.84	1.33	2.98	18.44	0.32	OK	0.74	0.153	OK	0.35
0.9D+1.6WL6	7.20	1.30	2.77	1.30	2.92	18.02	0.31	OK	0.74	0.15	OK	0.34
0.9D+1.6WL7	11.40	2.05	4.39	2.05	4.62	28.53	0.49	OK	0.74	0.237	OK	0.54
0.9D+1.6WL8	10.86	1.95	4.18	1.95	4.40	27.18	0.47	OK	0.74	0.226	OK	0.51

Load Case	kn (Ø=0.75)	p =	A_{st}
Ultimate Conditions			
1.4D	0.54	0.001	215
1.4D+1.7(CL1)+0.5Lr	0.79	0.002	430
1.4D+1.7(CL2)+0.5Lr	0.76	0.002	430
1.4D+1.7(CL3)+0.5Lr	0.98	0.002	430
1.4D+1.7(CL4)+0.5Lr	0.89	0.002	430
1.4D+1.7(CL5)+0.5Lr	0.75	0.002	430
1.4D+1.7(CL6)+0.5Lr	0.8	0.002	430
1.4D+1.7(CL5)+0.5Lr	0.91	0.002	430
1.4D+1.7(CL6)+0.5Lr	0.96	0.002	430
1.2D+1.6Lr+1.0(CL1)	0.89	0.002	430
1.2D+1.6Lr+1.0(CL2)	0.87	0.002	430
1.2D+1.6Lr+1.0(CL3)	1	0.002	430
1.2D+1.6Lr+1.0(CL4)	0.95	0.002	430
1.2D+1.6Lr+1.0(CL5)	0.87	0.002	430
1.2D+1.6Lr+1.0(CL6)	0.89	0.002	430
1.2D+1.6Lr+1.0(CL7)	0.96	0.002	430
1.2D+1.6Lr+1.0(CL8)	0.99	0.002	430
1.2D + 1.6WL1+1.0L+0.5Lr	0.58	0.001	215
1.2D + 1.6WL2+1.0L+0.5Lr	0.58	0.001	215
1.2D + 1.6WL3+1.0L+0.5Lr	0.4	0.001	215
1.2D + 1.6WL4+1.0L+0.5Lr	0.33	0.001	215
1.2D + 1.6WL5+1.0L+0.5Lr	0.61	0.001	215
1.2D + 1.6WL6+1.0L+0.5Lr	0.6	0.001	215
1.2D + 1.6WL7+1.0L+0.5Lr	0.9	0.002	430
1.2D + 1.6WL8+1.0L+0.5Lr	0.87	0.002	430
0.9D+1.6WL1	0.4	0.001	215
0.9D+1.6WL2	0.4	0.001	215
0.9D+1.6WL3	0.41	0.001	215
0.9D+1.6WL4	0.34	0.001	215
0.9D+1.6WL5	0.47	0.001	215
0.9D+1.6WL6	0.45	0.001	215
0.9D+1.6WL7	0.72	0.002	430
0.9D+1.6WL8	0.69	0.002	430

Location	Mux	ϕ Kn =	Kn	ρ=	Ast	Minm. Ast	Bar Dia	Spacing	Maxm	Spacing
At bottom	2.85	0.75	1.00	0.002	430.00	387.00	12øT	263.02	300	200
At top	0.00	0.00	0.00	0.0018	351.00	351.00	12øT	322.21	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**12øT**	**200**	c/c	At Top	**12øT**	**200**	c/c
Along Z Direction :	At Bottom	**12øT**	**200**	c/c	At Top	**12øT**	**200**	c/c

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	06-05-2020	01-06-2020	25-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
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Outotec / L&T Revision No.:	0	1	2	3
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3.4 Design of Foundation mkd. F4 (GRID D3 & A3)

	D3 (A3 SIMILAR)		
Load Combinations	FX (T)	FY (T)	FZ (T)
1 DL	0.9	6.1	0.3
2 LLROOF	1.0	2.9	0.2
3 CL1(L+XS)	0.2	2.0	0.1
4 CL2(L-XS)	0.1	2.0	0.1
5 CL3(R+XS)	0.5	4.8	0.2
6 CL4(R-XS)	0.2	4.8	0.2
7 CL5(L+ZT)	0.2	2.1	0.2
8 CL6(L-ZT)	0.2	1.8	0.0
9 CL7(R+ZT)	0.3	5.0	0.3
10 CL8(R-ZT)	0.3	4.6	0.1
21 WL1(+X)S	0.0	-0.8	-0.1
22 WL2(+X)P	0.1	-1.7	-0.1
23 WL3(-X)S	-1.4	-1.4	-0.1
24 WL4(-X)P	-1.2	-2.3	-0.1
25 WL5(+Z)S	-0.1	2.0	1.2
26 WL6(+Z)P	0.0	0.7	1.0
27 WL7(-Z)S	-0.2	-4.9	-1.3
28 WL8(-Z)P	0.0	-5.4	-1.2

Strength Design Load Combinations	D3 (A3 SIMILAR)		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4D	1.28	8.59	0.40
1.4D+1.7(CL1)+0.5Lr	2.23	13.45	0.62
1.4D+1.7(CL2)+0.5Lr	2.06	13.38	0.63
1.4D+1.7(CL3)+0.5Lr	2.69	18.22	0.86
1.4D+1.7(CL4)+0.5Lr	2.11	18.19	0.87
1.4D+1.7(CL5)+0.5Lr	2.14	13.71	0.76
1.4D+1.7(CL6)+0.5Lr	2.14	13.13	0.50
1.4D+1.7(CL5)+0.5Lr	2.40	18.49	0.99
1.4D+1.7(CL6)+0.5Lr	2.40	17.91	0.73
1.2D+1.6Lr+1.0(CL1)	3.02	14.06	0.67
1.2D+1.6Lr+1.0(CL2)	2.92	14.02	0.68
1.2D+1.6Lr+1.0(CL3)	3.29	16.87	0.81
1.2D+1.6Lr+1.0(CL4)	2.95	16.85	0.82
1.2D+1.6Lr+1.0(CL5)	2.97	14.22	0.75
1.2D+1.6Lr+1.0(CL6)	2.97	13.87	0.60
1.2D+1.6Lr+1.0(CL7)	3.12	17.03	0.89
1.2D+1.6Lr+1.0(CL8)	3.12	16.69	0.74
1.2D + 1.6WL1+1.0L+0.5Lr	1.65	7.59	0.32
1.2D + 1.6WL2+1.0L+0.5Lr	1.85	6.16	0.23
1.2D + 1.6WL3+1.0L+0.5Lr	-0.55	6.52	0.31
1.2D + 1.6WL4+1.0L+0.5Lr	-0.34	5.18	0.23
1.2D + 1.6WL5+1.0L+0.5Lr	1.49	12.10	2.32
1.2D + 1.6WL6+1.0L+0.5Lr	1.61	9.97	2.01
1.2D + 1.6WL7+1.0L+0.5Lr	1.34	1.05	-1.69
1.2D + 1.6WL8+1.0L+0.5Lr	1.63	0.14	-1.56
0.9D+1.6WL1	0.85	4.28	0.16
0.9D+1.6WL2	1.06	2.85	0.07
0.9D+1.6WL3	-1.34	3.21	0.15
0.9D+1.6WL4	-1.14	1.87	0.07
0.9D+1.6WL5	0.69	8.79	2.15
0.9D+1.6WL6	0.81	6.66	1.84
0.9D+1.6WL7	0.54	-2.26	-1.85
0.9D+1.6WL8	0.83	-3.17	-1.72

Global Project ID	900707
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Ma'aden Revision No:	A01	C01	D01
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DESIGN OF FOUNDATION MKD. F4

	About-X	About-Z
Size of column :	0.55m	0.55m
Size of fdn. raft :	1.75m	1.75m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 t/m^3	

SYMBOLS AND NOTATIONS

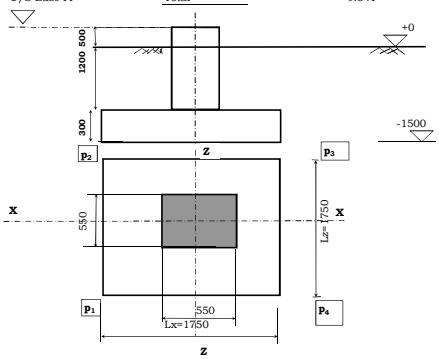
Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W
Crane Load	CL

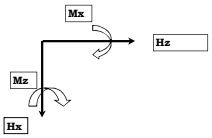
Calculation of Weight of Foundation :

Weight of raft & Col.= (1.75*1.75*0.3*2.5)+(0.55*0.55*1.7*2.5) = 3.58T

Weight of soil = ((1.75*1.75)-(0.55*0.55))*1.2*1.8 = 5.96T

U/S Base Pl Total = 9.54T





Design Input:

Characteristic strength of concrete, f'$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	20mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	20mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 215mm = 0.215m

Effective depth of slab along Z direction = 195mm = 0.195m

Clear projection along Z direction = 0.6m

Clear projection along X direction = 0.6m

WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+Lr	1.96	9.08	0.44	0.0	0.00	18.62	1.96	0.88	0.44	3.93
DL+CL1	1.17	8.13	0.37	0.0	0.00	17.67	1.17	0.74	0.37	2.33
DL+CL2	1.06	8.09	0.38	0.0	0.00	17.63	1.06	0.76	0.38	2.13
DL+CL3	1.43	10.93	0.51	0.0	0.00	20.47	1.43	1.02	0.51	2.87
DL+CL4	1.09	10.91	0.52	0.0	0.00	20.45	1.09	1.03	0.52	2.19
DL+CL5	1.12	8.28	0.45	0.0	0.00	17.82	1.12	0.90	0.45	2.23
DL+CL6	1.11	7.94	0.30	0.0	0.00	17.48	1.11	0.59	0.30	2.23
DL+CL7	1.26	11.10	0.59	0.0	0.00	20.64	1.26	1.18	0.59	2.53
DL+CL8	1.26	10.75	0.44	0.0	0.00	20.29	1.26	0.87	0.44	2.53
DL+0.75*CL1+0.75*Lr	1.89	9.84	0.46	0.0	0.00	19.38	1.89	0.93	0.46	3.78
DL+0.75*CL2+0.75*Lr	1.81	9.81	0.47	0.0	0.00	19.35	1.81	0.94	0.47	3.62
DL+0.75*CL3+0.75*Lr	2.09	11.94	0.57	0.0	0.00	21.48	2.09	1.14	0.57	4.18
DL+0.75*CL4+0.75*Lr	1.83	11.93	0.57	0.0	0.00	21.47	1.83	1.15	0.57	3.67
DL+0.75*CL5+0.75*Lr	1.85	9.95	0.52	0.0	0.00	19.49	1.85	1.05	0.52	3.70
DL+0.75*CL6+0.75*Lr	1.85	9.69	0.41	0.0	0.00	19.23	1.85	0.82	0.41	3.70
DL+0.75*CL7+0.75*Lr	1.96	12.06	0.63	0.0	0.00	21.60	1.96	1.26	0.63	3.93
DL+0.75*CL8+0.75*Lr	1.96	11.81	0.51	0.0	0.00	21.35	1.96	1.03	0.51	3.92
DL+0.75*Lr+0.75*WL1	1.71	7.76	0.36	0.0	0.00	17.30	1.71	0.71	0.36	3.42
DL+0.75*Lr+0.75*WL2	1.81	7.09	0.31	0.0	0.00	16.63	1.81	0.63	0.31	3.62
DL+0.75*Lr+0.75*WL3	0.68	7.26	0.35	0.0	0.00	16.80	0.68	0.70	0.35	1.37
DL+0.75*Lr+0.75*WL4	0.78	6.63	0.31	0.0	0.00	16.17	0.78	0.62	0.31	1.56
DL+0.75*Lr+0.75*WL5	1.64	9.87	1.29	0.0	0.00	19.41	1.64	2.58	1.29	3.28
DL+0.75*Lr+0.75*WL6	1.69	8.87	1.15	0.0	0.00	18.41	1.69	2.29	1.15	3.39
DL+0.75*Lr+0.75*WL7	1.57	4.69	-0.59	0.0	0.00	14.23	1.57	1.17	-0.59	3.14
DL+0.75*Lr+0.75*WL8	1.70	4.27	-0.53	0.0	0.00	13.81	1.70	1.05	-0.53	3.41
0.6*DL+WL1	0.56	2.91	0.11	0.0	0.00	12.45	0.56	0.22	0.11	1.13
0.6*DL+WL2	0.69	2.01	0.05	0.0	0.00	11.55	0.69	0.11	0.05	1.39
0.6*DL+WL3	-0.81	2.23	0.10	0.0	0.00	11.77	-0.81	0.21	0.10	1.61
0.6*DL+WL4	-0.68	1.40	0.05	0.0	0.00	10.94	-0.68	0.10	0.05	1.35
0.6*DL+WL5	0.47	5.72	1.36	0.0	0.00	15.26	0.47	2.71	1.36	0.93
0.6*DL+WL6	0.54	4.39	1.16	0.0	0.00	13.93	0.54	2.33	1.16	1.08
0.6*DL+WL7	0.37	-1.18	-1.15	0.0	0.00	8.36	0.37	2.29	-1.15	0.75
0.6*DL+WL8	0.55	-1.75	-1.07	0.0	0.00	7.79	0.55	2.13	-1.07	1.11

Check for overturning and sliding:

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
Load Case	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+Lr	18.62	1.96	3.93	0.44	0.88	4.15	18.48	3.79	16.90
DL+CL1	17.67	1.17	2.33	0.37	0.74	6.63	20.94	6.06	19.14
DL+CL2	17.63	1.06	2.13	0.38	0.76	7.25	20.39	6.63	18.64
DL+CL3	20.47	1.43	2.87	0.51	1.02	6.25	17.62	5.71	16.11
DL+CL4	20.45	1.09	2.19	0.52	1.03	8.18	17.34	7.47	15.85
DL+CL5	17.82	1.12	2.23	0.45	0.90	6.99	17.33	6.39	15.85
DL+CL6	17.48	1.11	2.23	0.30	0.59	6.86	25.80	6.28	23.59
DL+CL7	20.64	1.26	2.53	0.59	1.18	7.14	15.33	6.53	14.02
DL+CL8	20.29	1.26	2.53	0.44	0.87	7.03	20.39	6.42	18.64
DL+0.75*CL1+0.75*Lr	19.38	1.89	3.78	0.46	0.93	4.49	18.25	4.10	16.69
DL+0.75*CL2+0.75*Lr	19.35	1.81	3.62	0.47	0.94	4.67	17.96	4.27	16.42
DL+0.75*CL3+0.75*Lr	21.48	2.09	4.18	0.57	1.14	4.50	16.52	4.11	15.11
DL+0.75*CL4+0.75*Lr	21.47	1.83	3.67	0.57	1.15	5.12	16.34	4.68	14.94
DL+0.75*CL5+0.75*Lr	19.49	1.85	3.70	0.52	1.05	4.61	16.25	4.21	14.85
DL+0.75*CL6+0.75*Lr	19.23	1.85	3.70	0.41	0.82	4.55	20.53	4.16	18.77
DL+0.75*CL7+0.75*Lr	21.60	1.96	3.93	0.63	1.26	4.82	15.02	4.40	13.73
DL+0.75*CL8+0.75*Lr	21.35	1.96	3.92	0.51	1.03	4.76	18.16	4.35	16.61
DL+0.75*Lr+0.75*WL1	17.30	1.71	3.42	0.36	0.71	4.42	21.31	4.04	19.48
DL+0.75*Lr+0.75*WL2	16.63	1.81	3.62	0.31	0.63	4.02	23.22	3.68	21.23
DL+0.75*Lr+0.75*WL3	16.80	0.68	1.37	0.35	0.70	10.74	20.92	9.82	19.12
DL+0.75*Lr+0.75*WL4	16.17	0.78	1.56	0.31	0.62	9.05	22.65	8.27	20.71
DL+0.75*Lr+0.75*WL5	19.41	1.64	3.28	1.29	2.58	5.18	6.58	4.74	6.02
DL+0.75*Lr+0.75*WL6	18.41	1.69	3.39	1.15	2.29	4.76	7.03	4.35	6.43
DL+0.75*Lr+0.75*WL7	14.23	1.57	3.14	-0.59	1.17	3.97	10.63	3.63	9.72
DL+0.75*Lr+0.75*WL8	13.81	1.70	3.41	-0.53	1.05	3.54	11.48	3.24	10.50
0.6*DL+WL1	12.45	0.56	1.13	0.11	0.22	9.65	49.86	8.82	45.59
0.6*DL+WL2	11.55	0.69	1.39	0.05	0.11	7.28	94.79	6.65	86.67
0.6*DL+WL3	11.77	-0.81	1.61	0.10	0.21	6.39	49.53	5.84	45.28
0.6*DL+WL4	10.94	-0.68	1.35	0.05	0.10	7.08	92.03	6.47	84.14
0.6*DL+WL5	15.26	0.47	0.93	1.36	2.71	14.31	4.93	13.08	4.50
0.6*DL+WL6	13.93	0.54	1.08	1.16	2.33	11.30	5.24	10.33	4.79
0.6*DL+WL7	8.36	0.37	0.75	-1.15	2.29	9.80	3.19	8.96	2.92
0.6*DL+WL8	7.79	0.55	1.11	-1.07	2.13	6.15	3.20	5.63	2.92

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-05-2020	01-06-2020	25-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	06-05-2020	01-06-2020	25-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

ULTIMATE STRESS DESIGN	Load at U/S Of Base Pit					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4D	1.28	8.59	0.40	0.0	0.00	18.127	0.80	2.57
1.4D+1.7(CL1)+0.5Lr	2.23	13.45	0.62	0.0	0.00	22.991	1.24	4.46
1.4D+1.7(CL2)+0.5Lr	2.06	13.38	0.63	0.0	0.00	22.922	1.27	4.11
1.4D+1.7(CL3)+0.5Lr	2.69	18.22	0.86	0.0	0.00	27.758	1.71	5.37
1.4D+1.7(CL4)+0.5Lr	2.11	18.19	0.87	0.0	0.00	27.727	1.74	4.22
1.4D+1.7(CL5)+0.5Lr	2.14	13.71	0.76	0.0	0.00	23.247	1.51	4.29
1.4D+1.7(CL6)+0.5Lr	2.14	13.13	0.50	0.0	0.00	22.666	0.99	4.28
1.4D+1.7(CL5)+0.5Lr	2.40	18.49	0.99	0.0	0.00	28.034	1.99	4.80
1.4D+1.7(CL6)+0.5Lr	2.40	17.91	0.73	0.0	0.00	27.453	1.46	4.79
1.2D+1.6Lr+1.0(CL1)	3.02	14.06	0.67	0.0	0.00	23.605	1.35	6.05
1.2D+1.6Lr+1.0(CL2)	2.92	14.02	0.68	0.0	0.00	23.564	1.37	5.84
1.2D+1.6Lr+1.0(CL3)	3.29	16.87	0.81	0.0	0.00	26.409	1.63	6.58
1.2D+1.6Lr+1.0(CL4)	2.95	16.85	0.82	0.0	0.00	26.391	1.64	5.90
1.2D+1.6Lr+1.0(CL5)	2.97	14.22	0.75	0.0	0.00	23.756	1.51	5.95
1.2D+1.6Lr+1.0(CL6)	2.97	13.87	0.60	0.0	0.00	23.414	1.20	5.94
1.2D+1.6Lr+1.0(CL7)	3.12	17.03	0.89	0.0	0.00	26.571	1.79	6.25
1.2D+1.6Lr+1.0(CL8)	3.12	16.69	0.74	0.0	0.00	26.229	1.48	6.24
1.2D + 1.6WL1+1.0L+0.5Lr	1.65	7.59	0.32	0.0	0.00	17.132	0.64	3.29
1.2D + 1.6WL2+1.0L+0.5Lr	1.85	6.16	0.23	0.0	0.00	15.697	0.46	3.71
1.2D + 1.6WL3+1.0L+0.5Lr	-0.55	6.52	0.31	0.0	0.00	16.057	0.62	1.09
1.2D + 1.6WL4+1.0L+0.5Lr	-0.34	5.18	0.23	0.0	0.00	14.721	0.46	0.68
1.2D + 1.6WL5+1.0L+0.5Lr	1.49	12.10	2.32	0.0	0.00	21.641	4.63	2.98
1.2D + 1.6WL6+1.0L+0.5Lr	1.61	9.97	2.01	0.0	0.00	19.509	4.02	3.21
1.2D + 1.6WL7+1.0L+0.5Lr	1.34	1.05	-1.69	0.0	0.00	10.588	3.37	2.68
1.2D + 1.6WL8+1.0L+0.5Lr	1.63	0.14	-1.56	0.0	0.00	9.684	3.12	3.26
0.9D+1.6WL1	0.85	4.28	0.16	0.0	0.00	13.820	0.32	1.70
0.9D+1.6WL2	1.06	2.85	0.07	0.0	0.00	12.385	0.14	2.11
0.9D+1.6WL3	-1.34	3.21	0.15	0.0	0.00	12.745	0.30	2.69
0.9D+1.6WL4	-1.14	1.87	0.07	0.0	0.00	11.410	0.13	2.27
0.9D+1.6WL5	0.69	8.79	2.15	0.0	0.00	18.330	4.30	1.38
0.9D+1.6WL6	0.81	6.66	1.84	0.0	0.00	16.198	3.69	1.62
0.9D+1.6WL7	0.54	-2.26	-1.85	0.0	0.00	7.277	3.70	1.08
0.9D+1.6WL8	0.83	-3.17	-1.72	0.0	0.00	6.372	3.45	1.66

Check for Bearing Pressure

Load Combination & Calculation of Base pressure

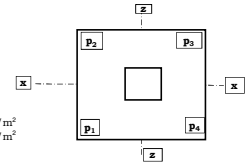
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 53.6 T/m^2 For WL/SL case = 66.1 T/m^2



Load Case	Design Load/moments			Base Pressure (T/m^2)								
	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2.Lz$	6.Mx/ $Lx.(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-p_3)	Remarks
DL+Lr	18.62	3.93	0.88	6.08	4.40	0.99	13.46	11.49	4.67	2.70	11.30	O.K
DL+CL1	17.67	2.33	0.74	5.77	2.61	0.83	11.21	9.55	5.99	4.33	9.05	O.K
DL+CL2	17.63	2.13	0.76	5.76	2.38	0.85	10.98	9.29	6.22	4.53	8.82	O.K
DL+CL3	20.47	2.87	1.02	6.69	3.21	1.14	13.03	10.76	6.61	4.34	10.87	O.K
DL+CL4	20.45	2.19	1.03	6.68	2.45	1.16	12.29	9.97	7.38	5.07	10.13	O.K
DL+CL5	17.82	2.23	0.90	5.82	2.50	1.01	11.32	9.31	6.33	4.31	9.16	O.K
DL+CL6	17.48	2.23	0.59	5.71	2.49	0.66	10.87	9.54	5.88	4.55	8.71	O.K
DL+CL7	20.64	2.53	1.18	6.74	2.83	1.32	12.89	10.25	7.22	4.59	10.73	O.K
DL+CL8	20.29	2.53	0.87	6.63	2.83	0.98	12.43	10.48	6.77	4.82	10.27	O.K
DL+0.75*CL1+0.75*Lr	19.38	3.78	0.93	6.33	4.23	1.04	13.60	11.52	5.14	3.06	11.44	O.K
DL+0.75*CL2+0.75*Lr	19.35	3.62	0.94	6.32	4.06	1.06	13.43	11.32	5.32	3.21	11.27	O.K
DL+0.75*CL3+0.75*Lr	21.48	4.18	1.14	7.01	4.68	1.27	14.97	12.42	5.61	3.06	12.81	O.K
DL+0.75*CL4+0.75*Lr	21.47	3.67	1.15	7.01	4.11	1.29	14.40	11.83	6.19	3.61	12.24	O.K
DL+0.75*CL5+0.75*Lr	19.49	3.70	1.05	6.36	4.14	1.18	13.68	11.33	5.40	3.05	11.52	O.K
DL+0.75*CL6+0.75*Lr	19.23	3.70	0.82	6.28	4.14	0.92	13.34	11.50	5.06	3.22	11.18	O.K
DL+0.75*CL7+0.75*Lr	21.60	3.93	1.26	7.05	4.39	1.41	14.86	12.04	6.07	3.25	12.70	O.K
DL+0.75*CL8+0.75*Lr	21.35	3.92	1.03	6.97	4.39	1.15	14.51	12.21	5.73	3.43	12.35	O.K
DL+0.75*Lr+0.75*WL1	17.30	3.42	0.71	5.65	3.83	0.80	12.28	10.69	4.61	3.02	10.12	O.K
DL+0.75*Lr+0.75*WL2	16.63	3.62	0.63	5.43	4.05	0.70	12.18	10.78	4.08	2.68	10.02	O.K
DL+0.75*Lr+0.75*WL3	16.80	1.37	0.70	5.48	1.53	0.79	9.80	8.23	6.74	5.17	7.64	O.K
DL+0.75*Lr+0.75*WL4	16.17	1.56	0.62	5.28	1.75	0.70	9.73	8.33	6.23	4.83	7.57	O.K
DL+0.75*Lr+0.75*WL5	19.41	3.28	2.58	6.34	3.67	2.89	14.90	9.12	7.56	1.78	12.74	O.K
DL+0.75*Lr+0.75*WL6	18.41	3.39	2.29	6.01	3.79	2.57	14.37	9.24	6.79	1.65	12.21	O.K
DL+0.75*Lr+0.75*WL7	14.23	3.14	1.17	4.65	3.51	1.31	11.47	8.85	4.45	1.82	9.31	O.K
DL+0.75*Lr+0.75*WL8	13.81	3.41	1.05	4.51	3.82	1.18	11.50	9.15	3.87	1.51	9.34	O.K
0.6*DL+WL1	12.45	1.13	0.22	4.06	1.26	0.24	7.57	7.08	5.04	4.56	5.41	O.K
0.6*DL+WL2	11.55	1.39	0.11	3.77	1.55	0.12	7.44	7.21	4.34	4.10	5.28	O.K
0.6*DL+WL3	11.77	1.61	0.21	3.84	1.80	0.23	7.88	7.42	4.27	3.81	5.72	O.K
0.6*DL+WL4	10.94	1.35	0.10	3.57	1.51	0.12	7.20	6.97	4.17	3.94	5.04	O.K
0.6*DL+WL5	15.26	0.93	2.71	4.98	1.04	3.04	11.06	4.99	8.98	2.90	8.90	O.K
0.6*DL+WL6	13.93	1.08	2.33	4.55	1.21	2.61	10.36	5.15	7.95	2.74	8.20	O.K
0.6*DL+WL7	8.36	0.75	2.29	2.73	0.84	2.56	8.13	3.00	6.46	1.33	5.97	O.K
0.6*DL+WL8	7.79	1.11	2.13	2.54	1.24	2.39	8.17	3.40	5.69	0.92	6.01	O.K

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-05-2020	01-06-2020	25-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	06-05-2020	01-06-2020	25-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

Calculation of Design Pressure

Load Combination & Calculation of Base pressure

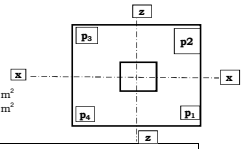
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 53.6 T/m^2 For WL/SL case = 66.1 T/m^2



Load Case	Design Load/moments			Base Pressure (T/m^2)							
	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-p_{th})
1.4D	18.1	2.57	0.80	5.92	2.87	0.90	11.69	9.90	5.94	4.15	8.99
1.4D+1.7(CL1)+0.5Lr	23.0	4.46	1.24	7.51	4.99	1.39	15.89	13.12	5.90	3.13	13.19
1.4D+1.7(CL2)+0.5Lr	22.9	4.11	1.27	7.48	4.60	1.42	15.51	12.67	6.30	3.46	12.81
1.4D+1.7(CL3)+0.5Lr	27.8	5.37	1.71	9.06	6.01	1.92	18.99	15.16	6.97	3.13	16.29
1.4D+1.7(CL4)+0.5Lr	27.7	4.22	1.74	9.05	4.72	1.95	17.72	13.83	8.28	4.39	15.02
1.4D+1.7(CL5)+0.5Lr	23.2	4.29	1.51	7.59	4.80	1.69	16.09	12.70	6.48	3.10	13.39
1.4D+1.7(CL6)+0.5Lr	22.7	4.28	0.99	7.40	4.80	1.11	15.31	13.09	5.71	3.50	12.61
1.4D+1.7(CL5)+0.5Lr	28.0	4.80	1.99	9.15	5.37	2.22	18.75	14.30	8.01	3.56	16.05
1.4D+1.7(CL6)+0.5Lr	27.5	4.79	1.46	8.96	5.37	1.64	17.97	14.69	7.24	3.96	15.27
1.2D+1.6Lr+1.0(CL1)	23.6	6.05	1.35	7.71	6.77	1.51	17.99	14.97	4.45	1.43	15.29
1.2D+1.6Lr+1.0(CL2)	23.6	5.84	1.37	7.69	6.54	1.53	17.76	14.71	4.68	1.62	15.06
1.2D+1.6Lr+1.0(CL3)	26.4	6.58	1.63	8.62	7.37	1.82	19.81	16.17	5.07	1.43	17.11
1.2D+1.6Lr+1.0(CL4)	26.4	5.90	1.64	8.62	6.61	1.84	19.07	15.39	5.84	2.17	16.37
1.2D+1.6Lr+1.0(CL5)	23.8	5.95	1.51	7.76	6.66	1.69	18.10	14.72	4.79	1.41	15.40
1.2D+1.6Lr+1.0(CL6)	23.4	5.94	1.20	7.65	6.65	1.35	17.65	14.95	4.34	1.65	14.95
1.2D+1.6Lr+1.0(CL7)	26.6	6.25	1.79	8.68	6.99	2.00	19.67	15.67	5.69	1.68	16.97
1.2D+1.6Lr+1.0(CL8)	26.2	6.24	1.48	8.56	6.99	1.66	19.21	15.90	5.23	1.92	16.51
1.2D + 1.6WL1+1.0L+0.5Lr	17.1	3.29	0.64	5.59	3.69	0.72	12.00	10.56	4.63	3.19	9.30
1.2D + 1.6WL2+1.0L+0.5Lr	15.7	3.71	0.46	5.13	4.15	0.52	11.79	10.76	3.49	2.46	9.09
1.2D + 1.6WL3+1.0L+0.5Lr	16.1	1.09	0.62	5.24	1.22	0.70	9.17	7.77	6.72	5.32	6.47
1.2D + 1.6WL4+1.0L+0.5Lr	14.7	0.68	0.46	4.81	0.76	0.51	8.08	7.05	6.56	5.54	5.38
1.2D + 1.6WL5+1.0L+0.5Lr	21.6	2.98	4.63	7.07	3.34	5.18	17.59	7.22	10.91	0.55	14.89
1.2D + 1.6WL6+1.0L+0.5Lr	19.5	3.21	4.02	6.37	3.60	4.50	16.46	7.47	9.27	0.28	13.76
1.2D + 1.6WL7+1.0L+0.5Lr	10.6	2.68	3.37	3.46	3.00	3.78	12.23	4.68	6.23	-1.32	9.53
1.2D + 1.6WL8+1.0L+0.5Lr	9.7	3.26	3.12	3.16	3.65	3.49	12.30	5.32	5.01	-1.98	9.60
0.9D+1.6WL1	13.8	1.70	0.32	4.51	1.90	0.35	8.76	8.06	4.97	4.26	6.06
0.9D+1.6WL2	12.4	2.11	0.14	4.04	2.36	0.15	8.56	8.26	3.83	3.53	5.86
0.9D+1.6WL3	12.7	2.69	0.30	4.16	3.01	0.33	9.51	8.84	3.49	2.82	6.81
0.9D+1.6WL4	11.4	2.27	0.13	3.73	2.54	0.15	8.42	8.12	3.33	3.03	5.72
0.9D+1.6WL5	18.3	1.38	4.30	5.99	1.55	4.82	14.35	4.71	11.26	1.62	11.65
0.9D+1.6WL6	16.2	1.62	3.69	5.29	1.81	4.13	13.23	4.97	9.61	1.35	10.53
0.9D+1.6WL7	7.3	1.08	3.70	2.38	1.21	4.14	9.73	1.45	7.30	-0.98	7.03
0.9D+1.6WL8	6.4	1.66	3.45	2.08	1.86	3.86	9.80	2.08	6.08	-1.64	7.10

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
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Date	06-05-2020	01-06-2020	25-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
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Approved	PRI	PRI	PRI	

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

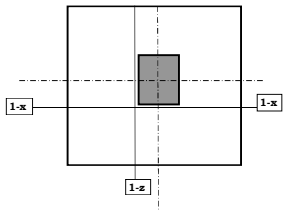
BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

$SF_{(oneway)}$ = Oneway shear force at a distance 'd' from the face of column in T/m.

$SF_{(punching)}$ = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.55+2*0.5*0.201)+(0.55+2*0.5*0.201))*2
= 2.98 m

Punching Area= ((0.55+2*0.5*0.195)*(0.55+2*0.5*0.195))
= 0.56 m^2 Punching shear stress to be checked



Downward pressure due to self weight of raft and soil = (0.3*2.5+1.2*1.8)
= 2.91 T/m^2

Area of footing, A_f= 3.06 m^2

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 42.82 T

Strength of Concrete in shear Vc1 = 124.28 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 132.28 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 114.59 T **Table 22.6.5.2_ (c)**

Design Vc= 114.59 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (42.82*10000)/(2980*195)
= 0.74 N/mm^2

No shear reinforcement reqd.

Load Case	p_{1x} (T)	$B.M_{1x}$ (T-m)	$S.F_{(Oneway)}$ (T)	$B.M_{1z}$ ™	$S.F_{(oneway)}$ (T)	$S.F_{(punching)}$ (T)	V_u/b_0d	Punching check (Remarks)	Allowable shear $(0.75*\sqrt{(fc)}/6$ (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4D	8.99	1.62	3.46	1.62	3.64	22.50	0.39	OK	0.74	0.187	OK	0.43
1.4D+1.7(CL1)+0.5Lr	13.19	2.37	5.08	2.37	5.34	33.01	0.57	OK	0.74	0.274	OK	0.62
1.4D+1.7(CL2)+0.5Lr	12.81	2.31	4.93	2.31	5.19	32.06	0.55	OK	0.74	0.266	OK	0.61
1.4D+1.7(CL3)+0.5Lr	16.29	2.93	6.27	2.93	6.60	40.77	0.7	OK	0.74	0.338	OK	0.77
1.4D+1.7(CL4)+0.5Lr	15.02	2.70	5.78	2.70	6.08	37.59	0.65	OK	0.74	0.312	OK	0.71
1.4D+1.7(CL5)+0.5Lr	13.39	2.41	5.16	2.41	5.42	33.51	0.58	OK	0.74	0.278	OK	0.63
1.4D+1.7(CL6)+0.5Lr	12.61	2.27	4.85	2.27	5.11	31.56	0.54	OK	0.74	0.262	OK	0.60
1.4D+1.7(CL5)+0.5Lr	16.05	2.89	6.18	2.89	6.50	40.17	0.69	OK	0.74	0.333	OK	0.76
1.4D+1.7(CL6)+0.5Lr	15.27	2.75	5.88	2.75	6.18	38.21	0.66	OK	0.74	0.317	OK	0.72
1.2D+1.6Lr+1.0(CL1)	15.29	2.75	5.89	2.75	6.19	38.26	0.66	OK	0.74	0.318	OK	0.72
1.2D+1.6Lr+1.0(CL2)	15.06	2.71	5.80	2.71	6.10	37.69	0.65	OK	0.74	0.313	OK	0.71
1.2D+1.6Lr+1.0(CL3)	17.11	3.08	6.59	3.08	6.93	42.82	0.74	OK	0.74	0.355	OK	0.81
1.2D+1.6Lr+1.0(CL4)	16.37	2.95	6.30	2.95	6.63	40.97	0.7	OK	0.74	0.34	OK	0.77
1.2D+1.6Lr+1.0(CL5)	15.40	2.77	5.93	2.77	6.24	38.54	0.66	OK	0.74	0.32	OK	0.73
1.2D+1.6Lr+1.0(CL6)	14.95	2.69	5.76	2.69	6.05	37.41	0.64	OK	0.74	0.311	OK	0.71
1.2D+1.6Lr+1.0(CL7)	16.97	3.05	6.53	3.05	6.87	42.47	0.73	OK	0.74	0.352	OK	0.80
1.2D+1.6Lr+1.0(CL8)	16.51	2.97	6.36	2.97	6.69	41.32	0.71	OK	0.74	0.343	OK	0.78
1.2D + 1.6WL1+1.0L+0.5Lr	9.30	1.67	3.58	1.67	3.77	23.27	0.4	OK	0.74	0.193	OK	0.44
1.2D + 1.6WL2+1.0L+0.5Lr	9.09	1.64	3.50	1.64	3.68	22.75	0.39	OK	0.74	0.189	OK	0.43
1.2D + 1.6WL3+1.0L+0.5Lr	6.47	1.16	2.49	1.16	2.62	16.19	0.28	OK	0.74	0.134	OK	0.31
1.2D + 1.6WL4+1.0L+0.5Lr	5.38	0.97	2.07	0.97	2.18	13.46	0.23	OK	0.74	0.112	OK	0.25
1.2D + 1.6WL5+1.0L+0.5Lr	14.89	2.68	5.73	2.68	6.03	37.26	0.64	OK	0.74	0.309	OK	0.70
1.2D + 1.6WL6+1.0L+0.5Lr	13.76	2.48	5.30	2.48	5.57	34.43	0.59	OK	0.74	0.286	OK	0.65
1.2D + 1.6WL7+1.0L+0.5Lr	9.53	1.72	3.67	1.72	3.86	23.85	0.41	OK	0.74	0.198	OK	0.45
1.2D + 1.6WL8+1.0L+0.5Lr	9.60	1.73	3.70	1.73	3.89	24.02	0.41	OK	0.74	0.199	OK	0.45
0.9D+1.6WL1	6.06	1.09	2.33	1.09	2.45	15.17	0.26	OK	0.74	0.126	OK	0.29
0.9D+1.6WL2	5.86	1.05	2.26	1.05	2.37	14.66	0.25	OK	0.74	0.122	OK	0.28
0.9D+1.6WL3	6.81	1.23	2.62	1.23	2.76	17.04	0.29	OK	0.74	0.141	OK	0.32
0.9D+1.6WL4	5.72	1.03	2.20	1.03	2.32	14.31	0.25	OK	0.74	0.119	OK	0.27
0.9D+1.6WL5	11.65	2.10	4.49	2.10	4.72	29.15	0.5	OK	0.74	0.242	OK	0.55
0.9D+1.6WL6	10.53	1.90	4.05	1.90	4.26	26.35	0.45	OK	0.74	0.219	OK	0.50
0.9D+1.6WL7	7.03	1.27	2.71	1.27	2.85	17.59	0.3	OK	0.74	0.146	OK	0.33
0.9D+1.6WL8	7.10	1.28	2.73	1.28	2.88	17.77	0.31	OK	0.74	0.147	OK	0.34

Load Case	kn (Ø=0.75)	p =	A_{st}
Ultimate Conditions			
1.4D	0.57	0.001	215
1.4D+1.7(CL1)+0.5Lr	0.83	0.002	430
1.4D+1.7(CL2)+0.5Lr	0.81	0.002	430
1.4D+1.7(CL3)+0.5Lr	1.03	0.002	430
1.4D+1.7(CL4)+0.5Lr	0.95	0.002	430
1.4D+1.7(CL5)+0.5Lr	0.85	0.002	430
1.4D+1.7(CL6)+0.5Lr	0.8	0.002	430
1.4D+1.7(CL5)+0.5Lr	1.01	0.002	430
1.4D+1.7(CL6)+0.5Lr	0.96	0.002	430
1.2D+1.6Lr+1.0(CL1)	0.97	0.002	430
1.2D+1.6Lr+1.0(CL2)	0.95	0.002	430
1.2D+1.6Lr+1.0(CL3)	1.08	0.003	645
1.2D+1.6Lr+1.0(CL4)	1.03	0.002	430
1.2D+1.6Lr+1.0(CL5)	0.97	0.002	430
1.2D+1.6Lr+1.0(CL6)	0.94	0.002	430
1.2D+1.6Lr+1.0(CL7)	1.07	0.003	645
1.2D+1.6Lr+1.0(CL8)	1.04	0.003	645
1.2D + 1.6WL1+1.0L+0.5Lr	0.59	0.001	215
1.2D + 1.6WL2+1.0L+0.5Lr	0.57	0.001	215
1.2D + 1.6WL3+1.0L+0.5Lr	0.41	0.001	215
1.2D + 1.6WL4+1.0L+0.5Lr	0.34	0.001	215
1.2D + 1.6WL5+1.0L+0.5Lr	0.94	0.002	430
1.2D + 1.6WL6+1.0L+0.5Lr	0.87	0.002	430
1.2D + 1.6WL7+1.0L+0.5Lr	0.6	0.001	215
1.2D + 1.6WL8+1.0L+0.5Lr	0.61	0.001	215
0.9D+1.6WL1	0.38	0.001	215
0.9D+1.6WL2	0.37	0.001	215
0.9D+1.6WL3	0.43	0.001	215
0.9D+1.6WL4	0.36	0.001	215
0.9D+1.6WL5	0.74	0.002	430
0.9D+1.6WL6	0.66	0.002	430
0.9D+1.6WL7	0.44	0.001	215
0.9D+1.6WL8	0.45	0.001	215

DESIGN OF ALL THE FOUNDATION ARE SAME
IN DRAWING ONLY ONE FOUNDATION DETAIL IS REPRESENTED
(Refer drawing number MD-216-49PR-EG-CV-GAD-1623)

Location	Mux	ϕ Kn =	Kn	ρ=	Ast	Minm. Ast	Bar Dia	Spacing	Maxm	Spacing
At bottom	3.08	0.81	1.08	0.003	645.00	387.00	12øT	175.34	300	200
At top	0.00	0.00	0.00	0.0018	351.00	351.00	12øT	322.21	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**12øT**	**200**	c/c	At Top	**12øT**	**200**	c/c
Along Z Direction :	At Bottom	**12øT**	**200**	c/c	At Top	**12øT**	**200**	c/c

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01			
Date	06-05-2020	01-06-2020	25-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI			

Outotec / L&T Revision No.:	0	1	2	3	
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Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI		

REFERENCES

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] ACI 318 : 2014 Building Code Requirement of Structural Concrete
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008
[6] WAREHOUSE DESIGN CALCULATION NO. MD-216-49PR-EG-ST-CAL-2420

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01			
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ATTACHMENT 1 Foundation Load Data for Workshop & Warehouse Building Columns

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2	3		
Date	06-05-2020	01-06-2020	25-10-2023				Date	06-05-2020	01-06-2020	25-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

1 DL	Dead load		
2 LLROOF	Live load on roof		
3 CL1(L+XS)	Crane load	Max Column concentration at A grid,Surge in direction in +X (from A grid to D grid)	
4 CL2(L-XS)	Crane load	Max Column concentration at A grid,Surge in direction in -X (from D grid to A grid)	
5 CL3(R+XS)	Crane load	Max Column concentration at D grid,Surge in direction in +X (from A grid to D grid)	
6 CL4(R-XS)	Crane load	Max Column concentration at D grid,Surge in direction in -X (from D grid to A grid)	
7 CL5(L+ZT)	Crane load	Max Column concentration at B grid,tractive in direction in +Z (from line-1 to line-6)	
8 CL6(L-ZT)	Crane load	Max Column concentration at B grid,tractive in direction in -Z (from line-6 to line-1)	
9 CL7(R+ZT)	Crane load	Max Column concentration at F grid,tractive in direction in +Z (from line-1 to line-6)	
10 CL8(R-ZT)	Crane load	Max Column concentration at F grid,tractive in direction in -Z (from line-6 to line-1)	
21 WL1(+X)S	Wind load	Wind load in direction of +X Suction(from A grid to D grid)	side wind
22 WL2(+X)P	Wind load	Wind load in direction of +X Pressure(from A grid to D grid)	side wind
23 WL3(-X)S	Wind load	Wind load in direction of -X Suction(from D grid to A grid)	side wind
24 WL4(-X)P	Wind load	Wind load in direction of -X Pressure(from D grid to A grid)	side wind
25 WL5(+Z)S	Wind load	Wind load in direction of +Z Suction(from line-1 to line-6)	gable wind
26 WL6(+Z)P	Wind load	Wind load in direction of +Z Pressure(from line-1 to line-6)	gable wind
27 WL7(-Z)S	Wind load	Wind load in direction of -Z Suction(from line-6 to line-1)	gable wind
28 WL8(-Z)P	Wind load	Wind load in direction of -Z Pressure(from line-6 to line-1)	gable wind

All crane and wind load cases are exclusive.

Sign convention

FY +ve sign indicates vertical load downward
FY -ve sign indicates vertical load upnward
FX +ve sign indicates horizonal load in across from grid-A to D
FX -ve sign indicates horizonal load in across from grid-D to A
FZ +ve sign indicates horizonal load in along from line 1 to 6
FZ -ve sign indicates horizonal load in along from line 6 to 1

LOAD DATA TABLE

		Load cases	Horizontal Fx Ton	Vertical Fy Ton	Horizontal Fz Ton
Line-1,4,5,6	D(1,4,5,6)	1 DL	1.0	6.1	0.0
		2 LLROOF	1.2	3.2	0.0
		3 CL1(L+XS)	0.2	2.0	0.0
		4 CL2(L-XS)	0.2	2.0	0.0
		5 CL3(R+XS)	0.5	4.8	0.0
		6 CL4(R-XS)	0.2	4.7	0.0
		7 CL5(L+ZT)	0.2	2.0	0.0
		8 CL6(L-ZT)	0.2	2.0	0.0
		9 CL7(R+ZT)	0.4	4.8	0.0
		10 CL8(R-ZT)	0.4	4.8	0.0
		21 WL1(+X)S	0.0	-0.8	0.0
		22 WL2(+X)P	0.0	-1.8	0.0
		23 WL3(-X)S	-1.4	-1.6	0.0
		24 WL4(-X)P	-1.3	-2.5	0.0
		25 WL5(+Z)S	-0.2	-1.5	0.0
		26 WL6(+Z)P	-0.1	-2.5	0.0
		27 WL7(-Z)S	-0.2	-1.5	0.0
		28 WL8(-Z)P	-0.1	-2.5	0.0
	A(1,4,5,6)	1 DL	-1.1	6.2	0.0
		2 LLROOF	-1.2	3.2	0.0
		3 CL1(L+XS)	-0.2	4.7	0.0
		4 CL2(L-XS)	-0.5	4.8	0.0
		5 CL3(R+XS)	-0.2	2.0	0.0
		6 CL4(R-XS)	-0.3	2.0	0.0
		7 CL5(L+ZT)	-0.3	4.8	0.0
		8 CL6(L-ZT)	-0.3	4.8	0.0
		9 CL7(R+ZT)	-0.2	2.0	0.0
		10 CL8(R-ZT)	-0.2	2.0	0.0
		21 WL1(+X)S	1.5	-1.7	0.0
		22 WL2(+X)P	1.4	-2.6	0.0
		23 WL3(-X)S	0.1	-0.9	0.0
		24 WL4(-X)P	0.0	-1.6	0.0
		25 WL5(+Z)S	0.2	-1.5	0.0
		26 WL6(+Z)P	0.2	-2.6	0.0
		27 WL7(-Z)S	0.2	-1.5	0.0
		28 WL8(-Z)P	0.1	-2.6	0.0
Line-2	D grid Line-2	1 DL	0.8	5.9	-0.2
		2 LLROOF	0.9	2.7	-0.1
		3 CL1(L+XS)	0.2	2.1	-0.1
		4 CL2(L-XS)	0.1	1.9	-0.1
		5 CL3(R+XS)	0.5	4.8	-0.2
		6 CL4(R-XS)	0.2	4.7	-0.2
		7 CL5(L+ZT)	0.2	1.8	0.0
		8 CL6(L-ZT)	0.2	2.2	-0.2
		9 CL7(R+ZT)	0.3	4.6	-0.1
		10 CL8(R-ZT)	0.3	4.9	-0.3
		21 WL1(+X)S	0.1	-0.6	0.0
		22 WL2(+X)P	0.2	-1.4	0.1
		23 WL3(-X)S	-1.3	-1.4	0.1
		24 WL4(-X)P	-1.1	-2.2	0.1
		25 WL5(+Z)S	0.0	-4.7	1.3
		26 WL6(+Z)P	0.1	-5.2	1.2
		27 WL7(-Z)S	-0.1	2.1	-1.2
		28 WL8(-Z)P	0.1	0.8	-1.0
	A grid Line-2	1 DL	-0.9	6.0	-0.2
		2 LLROOF	-0.9	2.7	-0.1
		3 CL1(L+XS)	-0.2	4.7	-0.2
		4 CL2(L-XS)	-0.5	4.8	-0.2
		5 CL3(R+XS)	-0.1	1.9	-0.1
		6 CL4(R-XS)	-0.3	2.1	-0.1
		7 CL5(L+ZT)	-0.3	4.6	-0.1
		8 CL6(L-ZT)	-0.3	4.9	-0.3
		9 CL7(R+ZT)	-0.2	1.8	0.0
		10 CL8(R-ZT)	-0.2	2.2	-0.2
		21 WL1(+X)S	1.4	-1.5	0.1
		22 WL2(+X)P	1.2	-2.3	0.1
		23 WL3(-X)S	0.0	-0.6	0.0
		24 WL4(-X)P	-0.2	-1.1	0.0
		25 WL5(+Z)S	0.1	-4.7	1.3
		26 WL6(+Z)P	0.0	-5.2	1.2
		27 WL7(-Z)S	0.0	2.1	-1.2
		28 WL8(-Z)P	-0.1	0.8	-1.0

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	06-05-2020	01-06-2020	25-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	06-05-2020	01-06-2020	25-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

Line-3	D grid Line-3	1 DL	0.9	6.1	0.3
		2 LLROOF	1.0	2.9	0.2
		3 CL1(L+XS)	0.2	2.0	0.1
		4 CL2(L-XS)	0.1	2.0	0.1
		5 CL3(R+XS)	0.5	4.8	0.2
		6 CL4(R-XS)	0.2	4.8	0.2
		7 CL5(L+ZT)	0.2	2.1	0.2
		8 CL6(L-ZT)	0.2	1.8	0.0
		9 CL7(R+ZT)	0.3	5.0	0.3
		10 CL8(R-ZT)	0.3	4.6	0.1
		21 WL1(+X)S	0.0	-0.8	-0.1
		22 WL2(+X)P	0.1	-1.7	-0.1
		23 WL3(-X)S	-1.4	-1.4	-0.1
		24 WL4(-X)P	-1.2	-2.3	-0.1
		25 WL5(+Z)S	-0.1	2.0	1.2
		26 WL6(+Z)P	0.0	0.7	1.0
		27 WL7(-Z)S	-0.2	-4.9	-1.3
		28 WL8(-Z)P	0.0	-5.4	-1.2
	A grid Line-3	1 DL	-1.0	6.2	0.3
		2 LLROOF	-1.1	3.0	0.2
		3 CL1(L+XS)	-0.2	4.8	0.2
		4 CL2(L-XS)	-0.5	4.8	0.2
		5 CL3(R+XS)	-0.1	2.0	0.1
		6 CL4(R-XS)	-0.3	2.0	0.1
		7 CL5(L+ZT)	-0.3	5.0	0.3
		8 CL6(L-ZT)	-0.3	4.6	0.1
		9 CL7(R+ZT)	-0.2	2.1	0.2
		10 CL8(R-ZT)	-0.2	1.8	0.0
		21 WL1(+X)S	1.4	-1.5	-0.1
		22 WL2(+X)P	1.3	-2.3	-0.1
		23 WL3(-X)S	0.0	-0.8	-0.1
		24 WL4(-X)P	-0.1	-1.4	-0.1
		25 WL5(+Z)S	0.1	2.0	1.2
		26 WL6(+Z)P	0.0	0.7	1.0
		27 WL7(-Z)S	0.1	-4.8	-1.3
		28 WL8(-Z)P	0.0	-5.4	-1.2
Gable post	gable post (B1,B6,C1,C6)	1 DL	0.0	4.4	0.0
		2 LLROOF	0.0	1.7	0.0
		3 CL1(L+XS)	0.0	-0.7	0.0
		4 CL2(L-XS)	0.0	0.0	0.0
		5 CL3(R+XS)	0.0	-0.1	0.0
		6 CL4(R-XS)	0.0	0.5	0.0
		7 CL5(L+ZT)	0.0	-0.3	0.0
		8 CL6(L-ZT)	0.0	-0.3	0.0
		9 CL7(R+ZT)	0.0	0.2	0.0
		10 CL8(R-ZT)	0.0	0.2	0.0
		21 WL1(+X)S	0.0	-1.4	0.5
		22 WL2(+X)P	0.0	-1.8	1.2
		23 WL3(-X)S	0.0	-0.1	0.5
		24 WL4(-X)P	0.0	-0.5	1.2
		25 WL5(+Z)S	0.0	-0.7	0.3
		26 WL6(+Z)P	0.0	-1.2	0.9
		27 WL7(-Z)S	0.0	-0.7	-1.2
		28 WL8(-Z)P	0.0	-1.6	-0.5